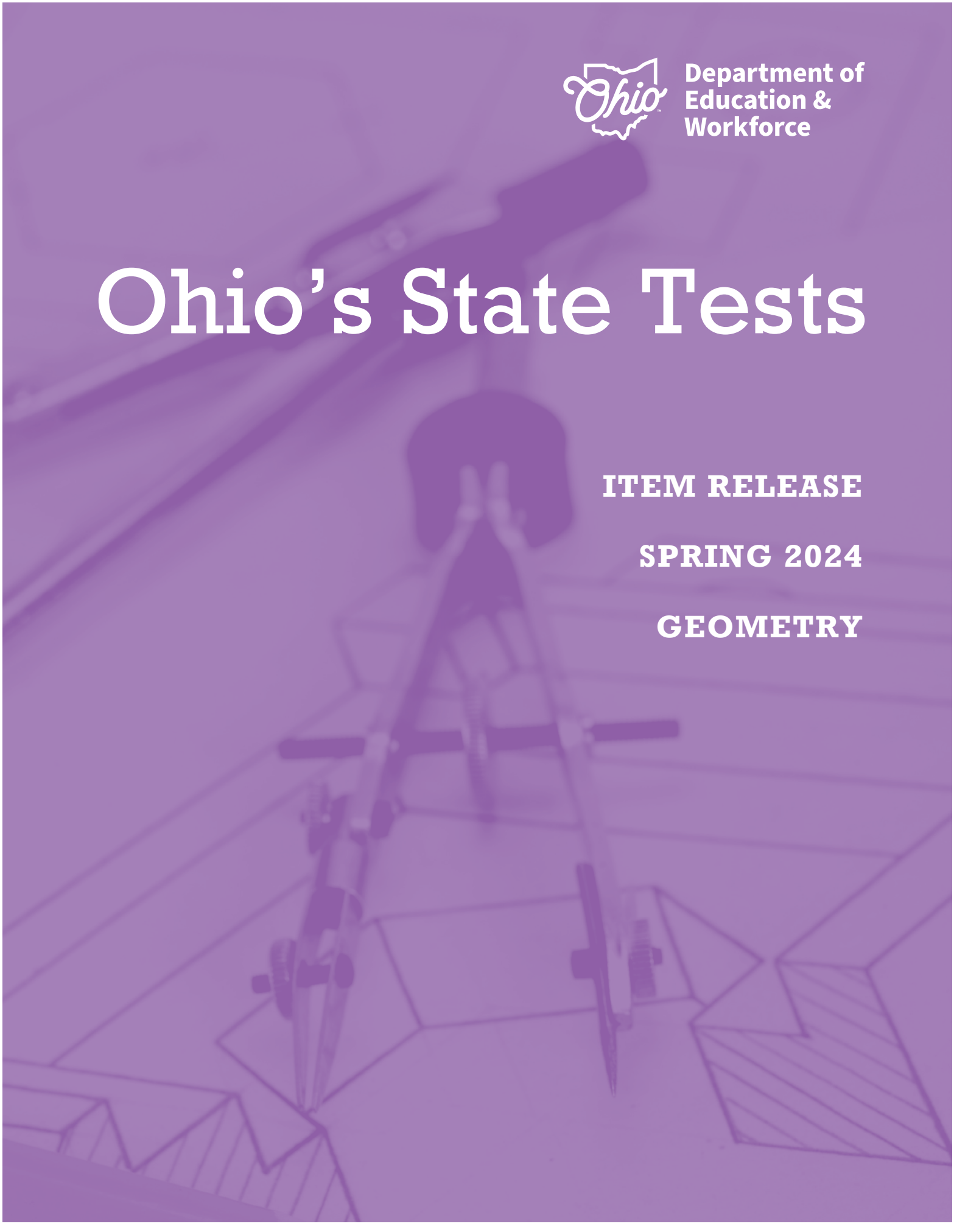


Department of Education & Workforce

Ohio's State Tests

ITEM RELEASE

SPRING 2024

GEOMETRY

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Geometry
Spring 2024 Item Release
Content Summary and Answer Key

Question No.*	Item Type	Content Cluster	Content Standard	Depth of Knowledge	Answer Key	Points
1	Multiple Choice Item	Find arc lengths and areas of sectors of circles.	*b.* Derive the formula for the area of a sector, and use it to solve problems. *(G.C.5)*	Level 1	C	1 point
3	Equation Item	Understand indepen-dence and conditional probability, and use them to interpret data.	Describe events as subsets of a sample space (the set of outcomes) using characteristics (or categories) of the outcomes, or as unions, intersections, or complements of other events (or, and, not). *(S.CP.1)*	Level 1	---	1 point
4	Multiple Choice Item	Prove and apply theorems both formally and informally involving similarity using a variety of methods.	Prove and apply theorems about triangles. Theorems include but are not restricted to the following: a line parallel to one side of a triangle divides the other two proportionally, and conversely; the Pythagorean Theorem proved using triangle similarity. *(G.SRT.4)*	Level 2	B	1 point
5	Inline Choice Item	Understand congruence in terms of rigid motions.	Use the definition of congruence in terms of rigid motions to show that two triangles are congruent if and only if corresponding pairs of sides and corresponding pairs of angles are congruent. *(G.CO.7)*	Level 3	---	1 point
11	Multiple Choice Item	Experiment with transform-ations in the plane.	Develop definitions of rotations, reflections, and translations in terms of angles, circles, perpendicular lines, parallel lines, and line segments. *(G.CO.4)*	Level 2	D	1 point

** The question numbers denote each question's place in the online test. Items are numbered sequentially on the practice site.*

Geometry
Spring 2024 Item Release
Content Summary and Answer Key

Question No.*	Item Type	Content Cluster	Content Standard	Depth of Knowledge	Answer Key	Points
12	Multi-Interaction Item	Explain volume formulas and use them to solve problems.	Give an informal argument for the formulas for the circumference of a circle, area of a circle, volume of a cylinder, pyramid, and cone. Use dissection arguments, Cavalieri's principle, and informal limit arguments. *(G.GMD.1)*	Level 3	---	2 points
18	Multiple Choice Item	Understand and apply theorems about circles.	Construct the inscribed and circumscribed circles of a triangle; prove and apply the property that opposite angles are supplementary for a quadrilateral inscribed in a circle. *(G.C.3)*	Level 1	D	1 point
19	Equation Item	Use coordinates to prove simple geometric theorems algebraically and to verify specific geometric statements.	Use coordinates to compute perimeters of polygons and areas of triangles and rectangles, e.g., using the distance formula. *(G.GPE.7)*	Level 2	---	1 point
20	Multi-Interaction Item	Understand the relationships between lengths, area, and volumes.	When figures are similar, understand and apply the fact that when a figure is scaled by a factor of k, the effect on lengths, areas, and volumes is that they are multiplied by k, k^2, and k^3, respectively. *(G.GMD.6)*	Level 2	---	2 points

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Geometry
Spring 2024 Item Release
Content Summary and Answer Key

Question No.*	Item Type	Content Cluster	Content Standard	Depth of Knowledge	Answer Key	Points
21	Equation Item	Use coordinates to prove simple geometric theorems algebraically and to verify specific geometric statements.	Use coordinates to prove simple geometric theorems algebraically and verify geometric relationships algebraically, including properties of special triangles, quadrilaterals, and circles. *For example, determine if a figure defined by four given points in the coordinate plane is a rectangle; determine if a specific point lies on a given circle. (G.GPE.4)*	Level 2	---	1 point
22	Equation Item	Define trigonometric ratios and solve problems involving right triangles.	Solve problems involving right triangles. *a.* Use trigonometric ratios and the Pythagorean Theorem to solve right triangles in applied problems if one of the two acute angles and a side length is given. *(G.SRT.8)*	Level 2	---	1 point
23	Multiple Choice Item	Prove geometric theorems both formally and informally using a variety of methods.	Prove and apply theorems about lines and angles. Theorems include but are not restricted to the following: vertical angles are congruent; when a transversal crosses parallel lines, alternate interior angles are congruent and corresponding angles are congruent; points on a perpendicular bisector of a line segment are exactly those equidistant from the segment's endpoints. *(G.CO.9)*	Level 3	D	1 point
26	Equation Item	Apply geometric concepts in modeling situations.	Apply concepts of density based on area and volume in modeling situations (*e.g., persons per square mile, BTUs per cubic foot*). *(G.MG.2)*	Level 1	---	1 point

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Geometry
Spring 2024 Item Release
Content Summary and Answer Key

Question No.*	Item Type	Content Cluster	Content Standard	Depth of Knowledge	Answer Key	Points
30	Equation Item	Understand and apply theorems about circles.	Identify and describe relationships among angles, radii, chords, tangents, and arcs and use them to solve problems. Include the relationship between central, inscribed, and circumscribed angles and their intercepted arcs; inscribed angles on a diameter are right angles; the radius of a circle is perpendicular to the tangent where the radius intersects the circle. *(G.C.2)*	Level 2	---	1 point
33	Inline Choice Item	Understand congruence in terms of rigid motions.	Explain how the criteria for triangle congruence (ASA, SAS, and SSS) follow from the definition of congruence in terms of rigid motions. *(G.CO.8)*	Level 1	---	1 point
35	Multiple Choice Item	Visualize relationships between two-dimensional and three-dimensional objects.	Identify the shapes of two-dimensional cross-sections of three-dimensional objects, and identify three-dimensional objects generated by rotations of two-dimensional objects. *(G.GMD.4)*	Level 1	B	1 point
36	Equation Item	Define trigonometric ratios and solve problems involving right triangles.	Understand that by similarity, side ratios in right triangles are properties of the angles in the triangle, leading to definitions of trigonometric ratios for acute angles. *(G.SRT.6)*	Level 2	---	1 point
43	Grid Item	Make geometric construct-ions.	Construct an equilateral triangle, a square, and a regular hexagon inscribed in a circle. *(G.CO.13)*	Level 2	---	1 point

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Geometry
Spring 2024 Item Release
Content Summary and Answer Key

Question No.*	Item Type	Content Cluster	Content Standard	Depth of Knowledge	Answer Key	Points
45	Inline Choice Item	Understand indepen-dence and conditional probability, and use them to interpret data.	Understand the conditional probability of A given B as $P(A \text{ and } B)/P(B)$, and interpret independence of A and B as saying that the conditional probability of A given B is the same as the probability of A, and the conditional probability of B given A is the same as the probability of B. *(S.CP.3)*	Level 2	---	1 point
53	Multiple Choice Item	Understand similarity in terms of similarity transform-ations.	Use the properties of similarity transformations to establish the AA criterion for two triangles to be similar. *(G.SRT.3)*	Level 1	A	1 point

** The question numbers denote each question's place in the online test.*
Items are numbered sequentially on the practice site.

Depth of Knowledge (DOK)

DOK refers to the complexity of thinking required to complete a task in a given item. Items with a DOK 1 designation focus on the recall of information, such as definitions and terms, and simple procedures. Items with a DOK 2 designation require students to make decisions, solve routine problems, perform calculations, or recognize patterns. Items with a DOK 3 designation feature higher-order cognitive tasks. These DOK 3 tasks include but are not limited to: critiquing a statement and forming a conclusion; explaining, justifying, or proving a statement; or approaching abstract, complex, open-ended, and non-routine problems. Each grade's blueprint contains information about the number of points of opportunity students will encounter at each DOK level.

Table 1: *Math Descriptors* – Applying Depth of Knowledge Levels for Mathematics (Webb, 2002) & NAEP 2002 Mathematics Levels of Complexity
(M. Petit, Center for Assessment 2003, K. Hess, Center for Assessment, updated 2006)

Level 1 Recall	Level 2 Skills/Concepts	Level 3 Strategic Thinking	Level 4* Extended Thinking
a. Recall, observe, or recognize a fact, definition, term, or property b. Apply/compute a well-known algorithm (e.g., sum, quotient) c. Apply a formula d. Determine the area or perimeter of rectangles or triangles given a drawing and labels e. Identify a plane or three-dimensional figure f. Measure g. Perform a specified or routine procedure (e.g., apply rules for rounding) h. Evaluate an expression i. Solve a one-step word problem j. Retrieve information from a table or graph	a. Classify plane and three-dimensional figures b. Interpret information from a simple graph c. Use models to represent mathematical concepts d. Solve a routine problem requiring multiple steps/ decision points, or the application of multiple concepts e. Compare and/or contrast figures or statements f. Construct 2-dimensional patterns for 3-dimensional models, such as cylinders and cones g. Provide justifications for steps in a solution process h. Extend a pattern	a. Interpret information from a complex graph b. Explain thinking when more than one response is possible c. Make and/or justify conjectures d. Use evidence to develop logical arguments for a concept e. Use concepts to solve non-routine problems f. Perform a procedure with multiple steps and multiple decision points g. Generalize a pattern h. Describe, compare, and contrast solution methods i. Formulate a mathematical model for a complex situation j. Provide mathematical justifications	a. Relate mathematical concepts to other content areas b. Relate mathematical concepts to real-world applications in new situations c. Apply a mathematical model to illuminate a problem, situation d. Conduct a project that specifies a problem, identifies solution paths, solves the problem, and reports results e. Design a mathematical model to inform and solve a practical or abstract situation f. Develop generalizations of the results obtained and the strategies used and apply them to new problem situations

Table 1 *continued on next page.*

Level 1 Recall	Level 2 Skills/Concepts	Level 3 Strategic Thinking	Level 4* Extended Thinking
k. Recall, identify, or make conversions between and among representations or numbers (fractions, decimals, and percents), or within and between customary and metric measures l. Locate numbers on a number line, or points on a coordinate grid m. Solve linear equations n. Represent math relationships in words, pictures, or symbols o. Read, write, and compare decimals in scientific notation	i. Retrieve information from a table, graph, or figure and use it to solve a problem requiring multiple steps j. Translate between tables, graphs, words and symbolic notation k. Make direct translations between problem situations and symbolic notation l. Select a procedure according to criteria and perform it m. Specify and explain relationships between facts, terms, properties, or operations n. Compare, classify, organize, estimate, or order data	k. Solve a multiple-step problem and provide support with a mathematical explanation that justifies the answer l. Solve 2-step linear equations/ inequalities in one variable over the rational numbers, interpret solution(s) in the original context, and verify reasonableness of results m. Translate between a problem situation and symbolic notation that is not a direct translation n. Formulate an original problem, given a situation o. Analyze the similarities and differences between procedures p. Draw conclusion from observations or data, citing evidence	g. Apply one approach among many to solve problems h. Apply understanding in a novel way, providing an argument/ justification for the application ***NOTE:*** *Level 4 involves such things as complex restructuring of data or establishing and evaluating criteria to solve problems.*

Note: Ohio's State Tests only assess and measure DOK Levels 1 – 3 in grades K – 12. Level 4 is included in this table for informational purposes only.

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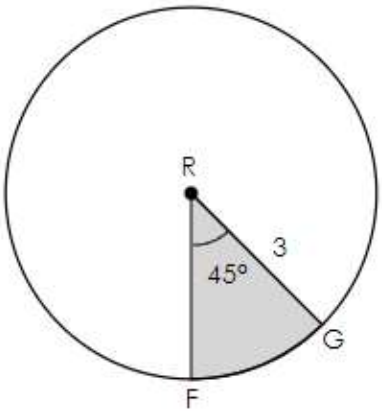
Geometry
Spring 2024 Item Release

Question 1

Question and Scoring Guidelines

Question 1

A circle is shown.



Which expression represents the shaded area formed by sector FRG?

Ⓐ $(3)^2\pi$

Ⓑ $\frac{(3)^2\pi}{360}$

Ⓒ $\frac{45}{360}(3)^2\pi$

Ⓓ $\frac{315}{360}(3)^2\pi$

Points Possible: 1

Content Cluster: Find arc lengths and areas of sectors of circles.

Content Standard: *b.* Derive the formula for the area of a sector, and use it to solve problems. *(G.C.5)*

Depth of Knowledge: Level 1

Modeling and Reasoning: No

Scoring Guidelines

Rationale for Option A: This is incorrect. The student may use the formula for the area of the whole circle rather than the formula for the area of a sector and does not use the proportional reasoning that the area of the sector is proportional to the measure of the central angle to find the area of the sector.

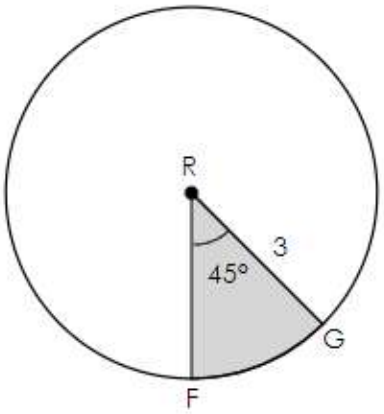
Rationale for Option B: This is incorrect. The student may realize the area of the sector can be calculated as a fraction of the area of the whole circle with a denominator of 360 rather than the formula for the area of a sector but does not use the proportional reasoning that the area of the sector is proportional to the measure of the central angle to find the area of the sector.

Rationale for Option C: **Key** –The student correctly identifies that the area of the sector is proportional to the measure of the central angle. The central angle of the sector measures 45°, so it is $\frac{45}{360}$ of the entire circle. Therefore, the sector's area is $\frac{45}{360}$ the area of the circle, or $\frac{45}{360}(3)^2\pi$.

Rationale for Option D: This is incorrect. The student may subtract the given angle of 45° from 360° to determine a central angle of 315°. The student then uses this angle in their proportional reasoning that the area of the sector is proportional to the measure of the central angle. Thus, the student finds the area of the sector by multiplying the area of the whole circle by the fraction created when dividing 315 by the total number of degrees in the circle, 360, finding the area of the sector formed by the central angle that is a reflex angle, and the major arc having the same endpoints as the radii.

Sample Response: 1 point

A circle is shown.



Which expression represents the shaded area formed by sector FRG?

Ⓐ $(3)^2\pi$

Ⓑ $\frac{(3)^2\pi}{360}$

● $\frac{45}{360}(3)^2\pi$

Ⓓ $\frac{315}{360}(3)^2\pi$

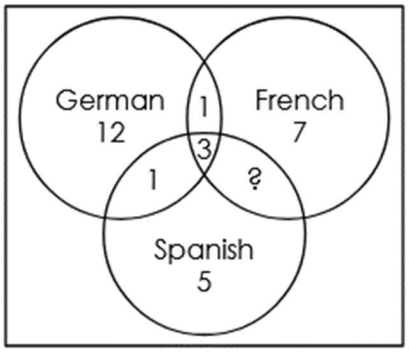
Geometry
Spring 2024 Item Release

Question 3

Question and Scoring Guidelines

Question 3

Alicia surveys 31 people about how many world languages they speak. She finds that 11 people speak Spanish, 17 people speak German, and 13 people speak French. Part of the data she collects is shown in the Venn diagram.



How many people does Alicia survey that speak both French and Spanish but not German?

Points Possible: 1

Content Cluster: Understand independence and conditional probability, and use them to interpret data.

Content Standard: Describe events as subsets of a sample space (the set of outcomes) using characteristics (or categories) of the outcomes, or as unions, intersections, or complements of other events (or, and, not). *(S.CP.1)*

Depth of Knowledge: Level 1

Modeling and Reasoning: No

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student correctly identifies that 2 people Alicia surveys speak both French and Spanish but not German, providing evidence of the ability to find the intersection of two events.

Exemplar Response

- 2
  or
- any equivalent value

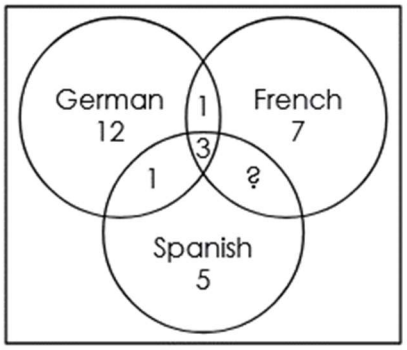
Geometry
Spring 2024 Item Release

Question 3

Sample Responses

Sample Response: 1 point

Alicia surveys 31 people about how many world languages they speak. She finds that 11 people speak Spanish, 17 people speak German, and 13 people speak French. Part of the data she collects is shown in the Venn diagram.



How many people does Alicia survey that speak both French and Spanish but not German?

2

Notes on Scoring

This response earns full credit (1 point) because it shows the correct value of the intersection of people who can speak French and Spanish but not German.

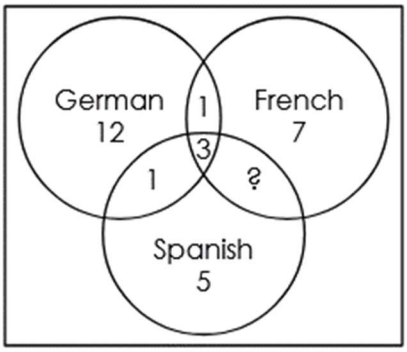
The student may use the fact that a total of 11 people surveyed speak Spanish along with the given information in the Venn diagram to find the number of people who speak French and Spanish but not German. The Venn diagram shows that 5 people only speak Spanish, 1 person speaks Spanish and German only, and 3 people speak all three languages. The equation $11 = 1 + 3 + 5 + x$ can be created to show the relationship of two different representations of the number of people surveyed who can speak Spanish, where x is the number of people who speak French and Spanish but not German. The student then solves this equation to find the unknown value.

$$\begin{aligned} 11 &= 1 + 3 + 5 + x \\ 11 &= 9 + x \\ 11 - 9 &= x \\ 2 &= x \end{aligned}$$

Thus, 2 people surveyed can speak both French and Spanish but not German.

Sample Response: 0 points

Alicia surveys 31 people about how many world languages they speak. She finds that 11 people speak Spanish, 17 people speak German, and 13 people speak French. Part of the data she collects is shown in the Venn diagram.



How many people does Alicia survey that speak both French and Spanish but not German?

12

Notes on Scoring

This response earns no credit (0 points) because it shows an incorrect value for the intersection of people who speak French and Spanish but not German.

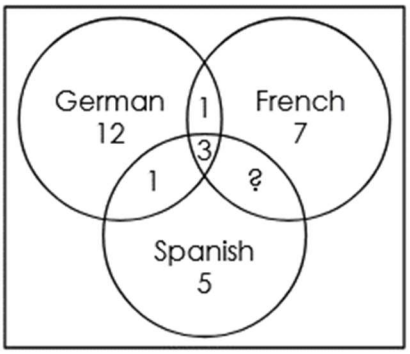
The student may correctly think that an equation needs to be created to relate the total number of people surveyed who can speak a language with the parts of the Venn diagram, but creates an equation that states that the total number of people who can speak Spanish or French (11 and 13, respectively) is equivalent to the sum of the number of people who can only speak French, 7, the number of people who only speak Spanish, 5, and the unknown number of people who speak French and Spanish but not German, x.

$$11 + 13 = 5 + 7 + x$$
$$24 = 12 + x$$
$$12 = x$$

The student correctly solves this equation and states that 12 people speak both French and Spanish but not German without considering that the overlapping values in the Venn diagram need to be represented in the equation as well since those numbers are part of the individual totals of people who can speak Spanish or can speak French.

Sample Response: 0 points

Alicia surveys 31 people about how many world languages they speak. She finds that 11 people speak Spanish, 17 people speak German, and 13 people speak French. Part of the data she collects is shown in the Venn diagram.



How many people does Alicia survey that speak both French and Spanish but not German?

1

Notes on Scoring

This response earns no credit (0 points) because it shows an incorrect value for the intersection of people who speak French and Spanish but not German.

The student may notice the number of students who speak both German and French but not Spanish and the number of students who speak both German and Spanish but not French are both 1. Since the number of students who speak both French and Spanish but not German is a similar case since only one of the three languages is not spoken, the student may think it must follow the pattern established by the other two, and be represented by 1, not considering the total number of students surveyed or the total number of students who can speak French or Spanish.

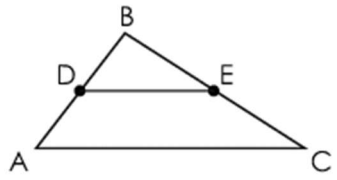
Geometry
Spring 2024 Item Release

Question 4

Question and Scoring Guidelines

Question 4

Triangle ABC and triangle DBE are shown. $\overline{DE}$ is parallel to $\overline{AC}$.



Marsha wants to prove that the side lengths of triangle DBE are proportional to those of triangle ABC.

An incomplete proof is given.

Statements	Reasons
1. $\overline{DE} \parallel \overline{AC}$	Given
2. $\angle BAC \cong \angle BDE$	Corresponding angles of parallel lines are congruent.
3. $\angle BCA \cong \angle BED$	Corresponding angles of parallel lines are congruent.

Which statement and reason would be the **best** next step in the proof?

Ⓐ AD = DB because D is the midpoint of $\overline{AB}$.

Ⓑ Triangle ABC is similar to triangle DBE by AA.

Ⓒ Triangle ABC is similar to triangle DBE by ASA.

Ⓓ $\frac{AB}{DB} = \frac{BC}{BE}$ because corresponding sides of similar triangles are in proportion.

Points Possible: 1

Content Cluster: Prove and apply theorems both formally and informally involving similarity using a variety of methods.

Content Standard: Prove and apply theorems about triangles. Theorems include but are not restricted to the following: a line parallel to one side of a triangle divides the other two proportionally, and conversely; the Pythagorean Theorem proved using triangle similarity. *(G.SRT.4)*

Depth of Knowledge: Level 2

Modeling and Reasoning: Yes

Scoring Guidelines

Rationale for Option A: This is incorrect. The student may note that because DE is parallel to AC, DE could connect midpoints, but does not consider that this parallelism guarantees only that the lengths of segments AD and DB are proportional to lengths of segments CE and EB, not that any must be congruent.

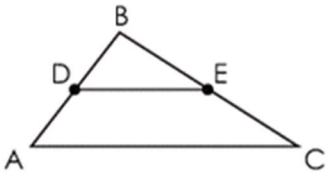
Rationale for Option B: **Key** – The student notes that the equal angles would prove similarity of two triangles by the AA postulate, which could then be used to prove proportionality of sides.

Rationale for Option C: This is incorrect. The student may think all parallel line segments are congruent, thinking that because two sides are given to be parallel and two pairs of angles are already proven to be congruent show $\triangle ABC$ is similar to $\triangle DBE$ by the ASA criterion.

Rationale for Option D: This is incorrect. The student may identify the conclusion of the proof rather than the next step without considering that the triangles must first be proven similar.

Sample Response: 1 point

Triangle ABC and triangle DBE are shown. $\overline{DE}$ is parallel to $\overline{AC}$.



Marsha wants to prove that the side lengths of triangle DBE are proportional to those of triangle ABC.

An incomplete proof is given.

Statements	Reasons
1. $\overline{DE} \parallel \overline{AC}$	Given
2. $\angle BAC \cong \angle BDE$	Corresponding angles of parallel lines are congruent.
3. $\angle BCA \cong \angle BED$	Corresponding angles of parallel lines are congruent.

Which statement and reason would be the **best** next step in the proof?

Ⓐ AD = DB because D is the midpoint of $\overline{AB}$.

● Triangle ABC is similar to triangle DBE by AA.

Ⓒ Triangle ABC is similar to triangle DBE by ASA.

Ⓓ $\frac{AB}{DB} = \frac{BC}{BE}$ because corresponding sides of similar triangles are in proportion.

Geometry
Spring 2024 Item Release

Question 5

Question and Scoring Guidelines

Question 5

Alfred correctly explains a correspondence between two triangles.

Select a word or phrase for each blank to complete Alfred's argument.

1. Suppose triangle ABC is congruent to triangle DEF.

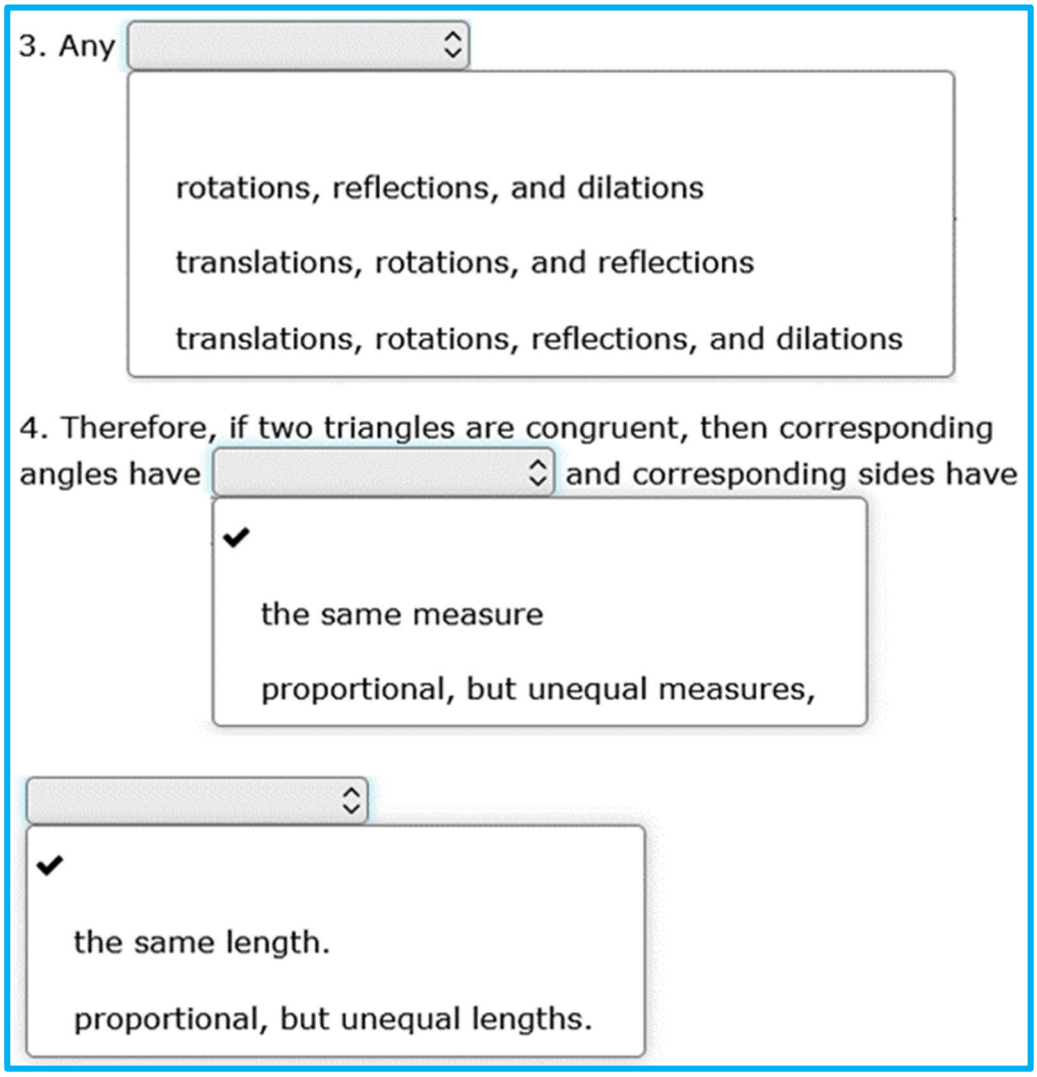
2. Then there is a series of rigid motions that maps triangle ABC onto triangle DEF.

3. Any [ ] of a triangle preserves the distances between every pair of corresponding points in the triangle.

4. Therefore, if two triangles are congruent, then corresponding angles have [ ] and corresponding sides have

[ ]

Drop Down Choices



Points Possible: 1

Content Cluster: Understand congruence in terms of rigid motions.

Content Standard: Use the definition of congruence in terms of rigid motions to show that two triangles are congruent if and only if corresponding pairs of sides and corresponding pairs of angles are congruent. *(G.CO.7)*

Depth of Knowledge: Level 3

Modeling and Reasoning: Yes

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student correctly completes the argument, providing evidence of the ability to use the definition of congruence in terms of rigid motions to show that two triangles are congruent if and only if corresponding pairs of sides and corresponding pairs of angles are congruent.

Exemplar Response

1. Suppose triangle ABC is congruent to triangle DEF.

2. Then there is a series of rigid motions that maps triangle ABC onto triangle DEF.

3. Any translations, rotations, and reflections of a triangle preserves the distances between every pair of corresponding points in the triangle.

4. Therefore, if two triangles are congruent, then corresponding angles have the same measure and corresponding sides have the same length.

Geometry
Spring 2024 Item Release

Question 5

Sample Responses

Sample Response: 1 point

Alfred correctly explains a correspondence between two triangles.

Select a word or phrase for each blank to complete Alfred's argument.

1. Suppose triangle ABC is congruent to triangle DEF.

2. Then there is a series of rigid motions that maps triangle ABC onto triangle DEF.

3. Any [translations, rotations, and reflections] of a triangle preserves the distances between every pair of corresponding points in the triangle.

4. Therefore, if two triangles are congruent, then corresponding angles have [the same measure] and corresponding sides have [the same length.]

Notes on Scoring

This response earns full credit (1 point) because it shows the correctly completed statements.

The student may first understand since rigid motions map ABC to DEF, dilations should not be included in the list as a dilation is not a rigid motion. The student also understands rigid motions preserve angle measures and sides lengths.

Sample Response: 0 points

Alfred correctly explains a correspondence between two triangles.

Select a word or phrase for each blank to complete Alfred's argument.

1. Suppose triangle ABC is congruent to triangle DEF.

2. Then there is a series of rigid motions that maps triangle ABC onto triangle DEF.

3. Any [translations, rotations, reflections, and dilations] of a triangle preserves the distances between every pair of corresponding points in the triangle.

4. Therefore, if two triangles are congruent, then corresponding angles have [the same measure] and corresponding sides have [the same length.]

Notes on Scoring

This response earns no credit (0 points) because it shows incorrectly completed statements.

The student may first think any transformation is a rigid motion, mistakenly including dilations in the list of rigid motions. The student also understands rigid motions preserve angle measures and sides lengths.

Sample Response: 0 points

Alfred correctly explains a correspondence between two triangles.

Select a word or phrase for each blank to complete Alfred's argument.

1. Suppose triangle ABC is congruent to triangle DEF.

2. Then there is a series of rigid motions that maps triangle ABC onto triangle DEF.

3. Any [rotations, reflections, and dilations] of a triangle preserves the distances between every pair of corresponding points in the triangle.

4. Therefore, if two triangles are congruent, then corresponding angles have [the same measure] and corresponding sides have [proportional, but unequal lengths.]

Notes on Scoring

This response earns no credit (0 points) because it shows incorrectly completed statements.

The student may confuse the definitions of congruency and similarity and thinks that translations and rotations proportionally change the length of the sides of a shape that is transformed like a dilation. The student then makes a conclusion defining similarity instead of congruency: that angle measures are preserved and the length of each side in the transformed shape change proportionally.

Geometry
Spring 2024 Item Release

Question 11

Question and Scoring Guidelines

Question 11

Luca translates quadrilateral PQRS x units right and y units up to create quadrilateral P′Q′R′S′, where x and y are not equal to zero.

Which statement **best** describes $\overline{PP'}$, $\overline{QQ'}$, $\overline{RR'}$, and $\overline{SS'}$?

Ⓐ Each line segment is the same distance from the other line segments.

Ⓑ Each line segment intersects the same point as the other line segments.

Ⓒ Each line segment is perpendicular to at least one of the other line segments.

Ⓓ Each line segment is either parallel to or collinear with the other line segments.

Points Possible: 1

Content Cluster: Experiment with transformations in the plane.

Content Standard: Develop definitions of rotations, reflections, and translations in terms of angles, circles, perpendicular lines, parallel lines, and line segments. *(G.CO.4)*

Depth of Knowledge: Level 2

Modeling and Reasoning: No

Scoring Guidelines

Rationale for Option A: This is incorrect. The student may not recognize that the distances between the line segments depends on the distance between the points in the initial quadrilateral, thus the line segments are not guaranteed to be the same distance apart.

Rationale for Option B: This is incorrect. The student may confuse a translation with a dilation about a point, say point T. In that case, $\overline{\mathrm{PP'}}$, $\overline{\mathrm{QQ'}}$, $\overline{\mathrm{RR'}}$, and $\overline{\mathrm{SS'}}$ would all intersect at point T.

Rationale for Option C: This is incorrect. The student may think that when each point is moved *x* units right and *y* units up, the slopes of the resulting line segments can vary, thinking that when both *x* and *y* are positive, the line segments have a slope of $\frac{y}{x}$ but when *x* or *y* is negative, the slope becomes $-\frac{x}{y}$. Thus, the student may think that some of the line segments are perpendicular.

Rationale for Option D: **Key** – The student understands that when each point is moved *x* units right and *y* units up, the resulting line segments, $\overline{\mathrm{PP'}}$, $\overline{\mathrm{QQ'}}$, $\overline{\mathrm{RR'}}$, and $\overline{\mathrm{SS'}}$ all have a slope of $\frac{y}{x}$, and that line segments with the same slope are either parallel or collinear.

Sample Response: 1 point

Luca translates quadrilateral PQRS x units right and y units up to create quadrilateral P′Q′R′S′, where x and y are not equal to zero.

Which statement **best** describes $\overline{PP'}$, $\overline{QQ'}$, $\overline{RR'}$, and $\overline{SS'}$?

- (A) Each line segment is the same distance from the other line segments.
- (B) Each line segment intersects the same point as the other line segments.
- (C) Each line segment is perpendicular to at least one of the other line segments.
- ● Each line segment is either parallel to or collinear with the other line segments.

Geometry
Spring 2024 Item Release

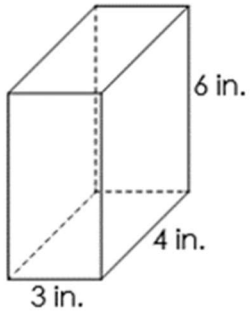
Question 12

Question and Scoring Guidelines

Question 12

This item has **two** parts.

The figure shows the dimensions, in inches (in.), of a wax candle.



The candle is melted and remolded in the shape of a right cylinder of height 4 inches so that the volume remains the same.

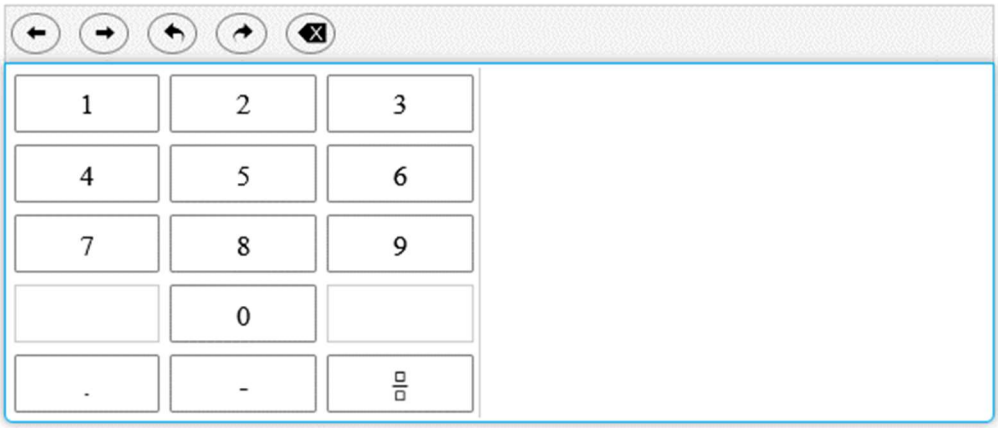
Part A. Complete the sentence to relate the dimensions of the two shapes.

As the volumes of the two shapes are the same and the height of the cylinder is equal to one of the side lengths of the prism, the [ ] of the cylindrical candle must be equal to the [ ] of the face of the prism with side lengths that are not equal to the height of the cylinder.

Part B. Complete the sentences to describe the two shapes.

The face of the prism with side lengths not equal to the height of the cylinder is [ ] inches by [ ] inches.

The radius, rounded to the nearest hundredth of an inch, of the base of the cylindrical candle is [ ] inches.



Drop Down Choices

As the volumes of the two shapes are the same and the height of the cylinder is equal to one of the side lengths of the prism, the [ ] of the cylindrical candle must be equal to

- ✔
- area of the base
- circumference of the base

the [ ] of the face of the prism with side

- ✔
- area
- perimeter

lengths that are not equal to the height of the cylinder.

Points Possible: 2

Content Cluster: Explain volume formulas and use them to solve problems.

Content Standard: Give an informal argument for the formulas for the circumference of a circle, area of a circle, volume of a cylinder, pyramid, and cone. Use dissection arguments, Cavalieri's principle, and informal limit arguments. *(G.GMD.1)*

Depth of Knowledge: Level 3

Modeling and Reasoning: Yes

Scoring Guidelines

For full credit (2 points), the student's response satisfies both bullets below.

For partial credit (1 point), the student's response satisfies one of the bullets below.

- The student correctly completes the statement that relates the dimensions of the two shapes, providing evidence of the ability to understand Cavalieri's principle.

- The student correctly identifies the radius of the base as 2.39 inches, providing evidence of the ability to use Cavalieri's principle to solve problems.

Exemplar Response

2 points

Part A. Complete the sentence to relate the dimensions of the two shapes.

As the volumes of the two shapes are the same and the height of the cylinder is equal to one of the side lengths of the prism, the [area of the base] of the cylindrical candle must be equal to the [area] of the face of the prism with side lengths that are not equal to the height of the cylinder.

Part B. Complete the sentences to describe the two shapes.

The face of the prism with side lengths not equal to the height of the cylinder is [6] inches by [3] inches.

The radius, rounded to the nearest hundredth of an inch, of the base of the cylindrical candle is [2.39] inches.

- Note: Any value from 2.39 to 2.394261, inclusive, is accepted for the radius, and the 3 and 6 can be in either box.

1 point

Part A. Complete the sentence to relate the dimensions of the two shapes.

As the volumes of the two shapes are the same and the height of the cylinder is equal to one of the side lengths of the prism, the [circumference of the base] of the cylindrical candle must be equal to the [perimeter] of the face of the prism with side lengths that are not equal to the height of the cylinder.

Part B. Complete the sentences to describe the two shapes.

The face of the prism with side lengths not equal to the height of the cylinder is [6] inches by [3] inches.

The radius, rounded to the nearest hundredth of an inch, of the base of the cylindrical candle is [2.39] inches.

Geometry
Spring 2024 Item Release

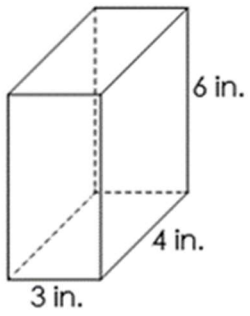
Question 12

Sample Responses

Sample Response: 2 points

This item has **two** parts.

The figure shows the dimensions, in inches (in.), of a wax candle.



The candle is melted and remolded in the shape of a right cylinder of height 4 inches so that the volume remains the same.

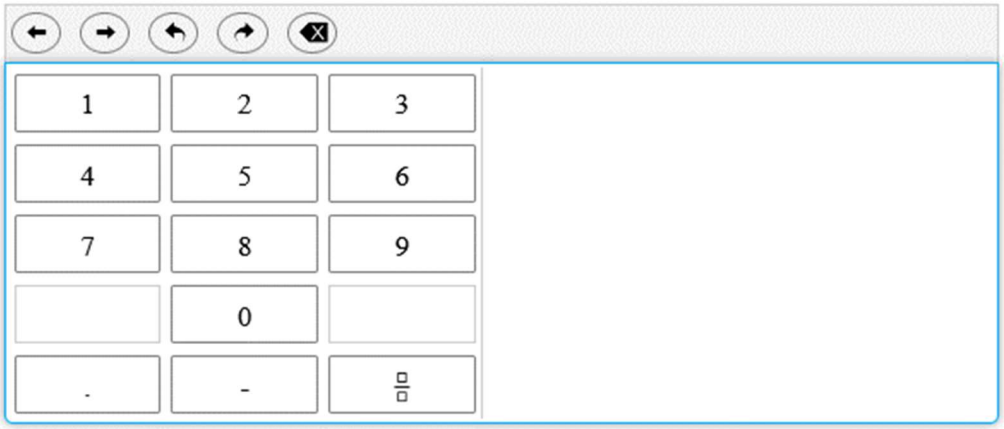
Part A. Complete the sentence to relate the dimensions of the two shapes.

As the volumes of the two shapes are the same and the height of the cylinder is equal to one of the side lengths of the prism, the [area of the base] of the cylindrical candle must be equal to the [area] of the face of the prism with side lengths that are not equal to the height of the cylinder.

Part B. Complete the sentences to describe the two shapes.

The face of the prism with side lengths not equal to the height of the cylinder is 6 inches by 3 inches.

The radius, rounded to the nearest hundredth of an inch, of the base of the cylindrical candle is 2.39 inches.



Notes on Scoring

This response earns full credit (2 points) because it shows correctly completed statements.

The student may use the volume equations to create the equality shown, with V_p representing the volume of the prism and V_c representing the volume of the cylindrical candle.

$$V_p = V_c$$
$$l \cdot w \cdot h = \pi \cdot r^2 \cdot h$$
$$l \cdot w = \pi \cdot r^2$$

Since the height of the prism and the cylinder are the same, the student can divide each side of the equation by *h* to remove it from the equation to find the area of the base of the cylindrical candle, $\pi \cdot r^2$, must be equal to the area of the face of the prism where both side lengths are not the height of the prism, *l* × *w*.

Since the height of the cylinder is 4 inches and the prism and cylinder have the same height, the student then identifies the remaining side lengths of the prism, 6 inches and 3 inches, are the values of *l* and *w*. To find the length of the radius, *r*, the student substitutes the values for *l* and *w* into the equation they used to answer Part A and solves it for *r*.

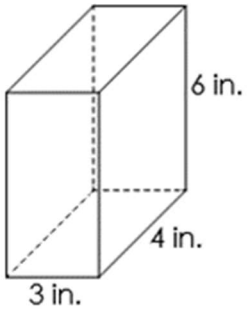
$$l \cdot w = \pi \cdot r^2$$
$$6 \cdot 3 = \pi \cdot r^2$$
$$18 = \pi \cdot r^2$$
$$\frac{18}{\pi} = r^2$$
$$\sqrt{\frac{18}{\pi}} = r$$
$$2.39 \approx r$$

Thus, the student finds the radius of the base of the cylindrical candle to be approximately 2.39 inches.

Sample Response: 2 points

This item has **two** parts.

The figure shows the dimensions, in inches (in.), of a wax candle.



The candle is melted and remolded in the shape of a right cylinder of height 4 inches so that the volume remains the same.

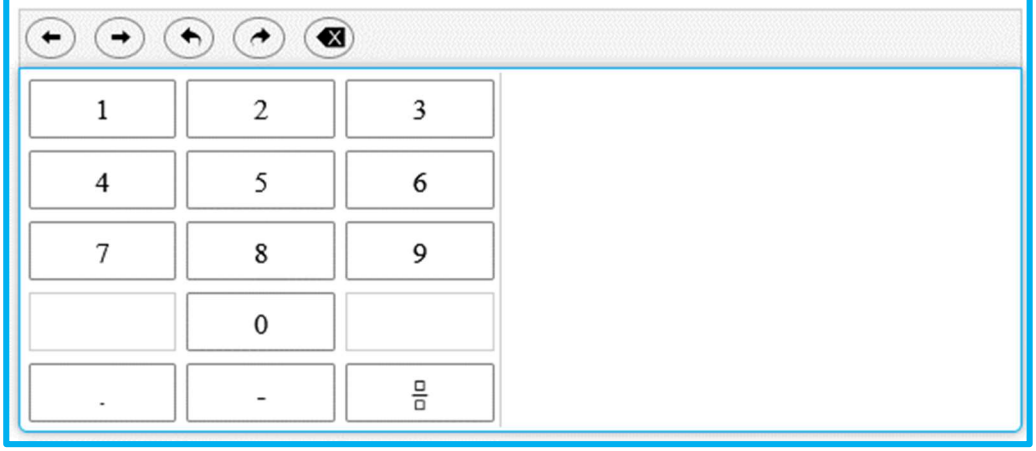
Part A. Complete the sentence to relate the dimensions of the two shapes.

As the volumes of the two shapes are the same and the height of the cylinder is equal to one of the side lengths of the prism, the [area of the base] of the cylindrical candle must be equal to the [area] of the face of the prism with side lengths that are not equal to the height of the cylinder.

Part B. Complete the sentences to describe the two shapes.

The face of the prism with side lengths not equal to the height of the cylinder is 6 inches by 3 inches.

The radius, rounded to the nearest hundredth of an inch, of the base of the cylindrical candle is 2.394 inches.



Notes on Scoring

This response earns full credit (2 points) because it shows correctly completed statements.

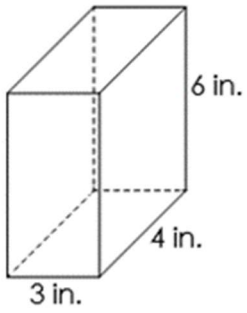
The student may understand that the base of the cylinder has the same area as the base of the prism when the height of the prism is 4 inches. The student then sets these areas equal to each other and solves for *r*, the length of the radius. The student gives a more precise length of the radius than requested in the item.

$$\begin{aligned} l \cdot w &= \pi \cdot r^2 \\ 6 \cdot 3 &= \pi \cdot r^2 \\ 18 &= \pi \cdot r^2 \\ \frac{18}{\pi} &= r^2 \\ \sqrt{\frac{18}{\pi}} &= r \\ 2.394 &\approx r \end{aligned}$$

Sample Response: 1 point

This item has **two** parts.

The figure shows the dimensions, in inches (in.), of a wax candle.



The candle is melted and remolded in the shape of a right cylinder of height 4 inches so that the volume remains the same.

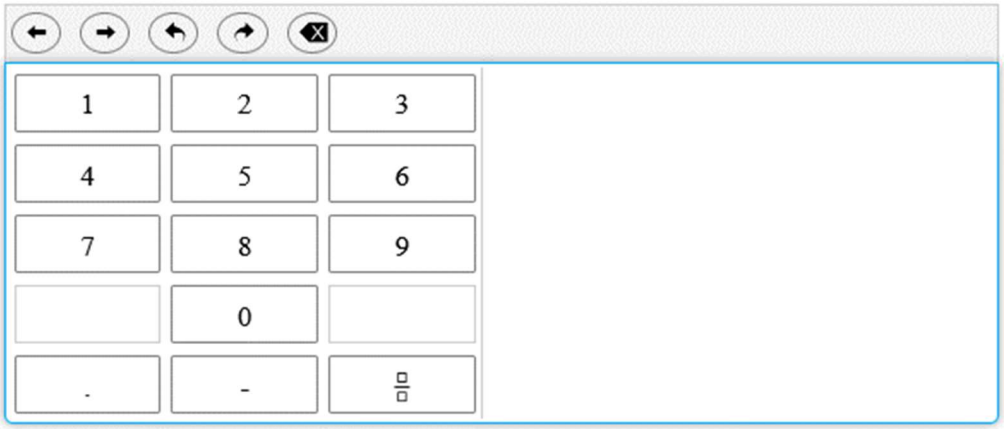
Part A. Complete the sentence to relate the dimensions of the two shapes.

As the volumes of the two shapes are the same and the height of the cylinder is equal to one of the side lengths of the prism, the [circumference of the base] of the cylindrical candle must be equal to the [perimeter] of the face of the prism with side lengths that are not equal to the height of the cylinder.

Part B. Complete the sentences to describe the two shapes.

The face of the prism with side lengths not equal to the height of the cylinder is 6 inches by 3 inches.

The radius, rounded to the nearest hundredth of an inch, of the base of the cylindrical candle is 2.39 inches.



Notes on Scoring

This response earns partial credit (1 point) because it shows an incorrect Part A and a correct Part B.

The student may attempt to create a relationship with the perimeter of the face of the prism and the circumference of the base of the cylindrical candle but does not consider they are only given information about the volume of each solid and the perimeter and circumference are not a part of either volume formula to create a valid relationship.

Since the height of the cylinder is 4 inches and the prism and cylinder have the same height, the student then identifies the remaining side lengths of the prism, 6 inches and 3 inches, are the length and width of the face of the prism. However, the student then confuses both the perimeter and circumference formulas with the formulas for area of a rectangle and area of a circle, respectively and uses the equation shown to find the radius of the cylinder's base.

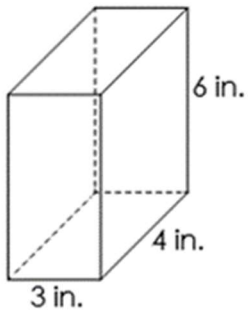
$$l \cdot w = \pi \cdot r^2$$
$$6 \cdot 3 = \pi \cdot r^2$$
$$2.39 \approx r$$

Thus, the student finds the radius of the base of the cylindrical candle to be approximately 2.39 inches.

Sample Response: 1 point

This item has **two** parts.

The figure shows the dimensions, in inches (in.), of a wax candle.



The candle is melted and remolded in the shape of a right cylinder of height 4 inches so that the volume remains the same.

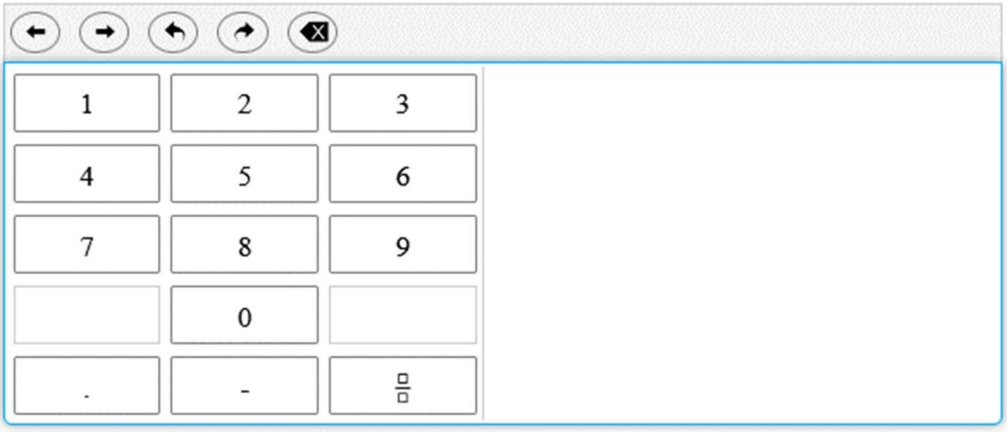
Part A. Complete the sentence to relate the dimensions of the two shapes.

As the volumes of the two shapes are the same and the height of the cylinder is equal to one of the side lengths of the prism, the [area of the base] of the cylindrical candle must be equal to the [area] of the face of the prism with side lengths that are not equal to the height of the cylinder.

Part B. Complete the sentences to describe the two shapes.

The face of the prism with side lengths not equal to the height of the cylinder is 6 inches by 3 inches.

The radius, rounded to the nearest hundredth of an inch, of the base of the cylindrical candle is 5.73 inches.



Notes on Scoring

This response earns partial credit (1 point) because it shows a correct Part A and an incorrect Part B.

The student may understand that the base of the cylinder has the same area as the base of the prism when the height of the prism is 4 inches. However, the student may have used incorrect formulas for the areas of the bases when calculating Part B. The student may have used the formula for the circumference of the base of the cylinder instead of using the formula for the area, and the student may have conflated the formulas for perimeter and area and used $2(l \times w)$ instead of just using $l \times w$ for the area of the base of the prism.

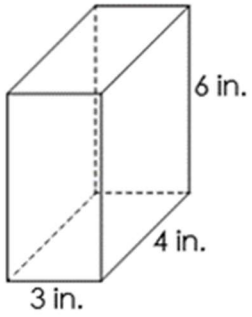
$$\begin{aligned} 2(6 \cdot 3) &= 2\pi r \\ 6 \cdot 3 &= \pi r \\ 18 &= \pi r \\ \frac{18}{\pi} &= r \end{aligned}$$

Thus, the student thinks the radius of the base of the cylindrical candle is approximately 5.73 inches.

Sample Response: 0 points

This item has **two** parts.

The figure shows the dimensions, in inches (in.), of a wax candle.



The candle is melted and remolded in the shape of a right cylinder of height 4 inches so that the volume remains the same.

Part A. Complete the sentence to relate the dimensions of the two shapes.

As the volumes of the two shapes are the same and the height of the cylinder is equal to one of the side lengths of the prism, the [circumference of the base] of the cylindrical candle must be equal to the [perimeter] of the face of the prism with side lengths that are not equal to the height of the cylinder.

Part B. Complete the sentences to describe the two shapes.

The face of the prism with side lengths not equal to the height of the cylinder is 6 inches by 3 inches.

The radius, rounded to the nearest hundredth of an inch, of the base of the cylindrical candle is 2.86 inches.



Notes on Scoring

This response earns no credit (0 points) because it shows incorrectly completed statements.

The student may attempt to create a relationship with the perimeter of the face of the prism and the circumference of the base of the cylindrical candle but does not consider they are only given information about the volume of each solid and the perimeter and circumference are not a part of either volume formula to create a valid relationship.

The student may then recognize that since the height of the cylinder is 4 inches and the prism and cylinder have the same height, the remaining lengths of the prism are 6 inches and 3 inches. Using this information and the assumption that the perimeter of the face of the prism and the circumference of the base of the cylinder are the same, the student creates the equation shown to find the radius of the cylinder.

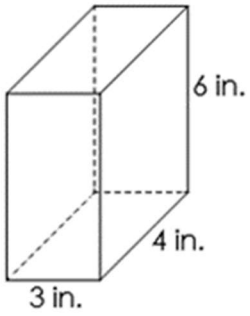
$$\begin{aligned} 2(3+6) &= 2\pi r \\ 2(9) &= 2\pi r \\ 9 &= \pi r \\ \frac{9}{\pi} &= r \end{aligned}$$

Thus, the student thinks the radius of the base of the cylindrical candle is approximately 2.86 inches, but since the perimeter of the base of the prism is not equal to the circumference of the base of the cylinder, the value of the radius is incorrect.

Sample Response: 0 points

This item has **two** parts.

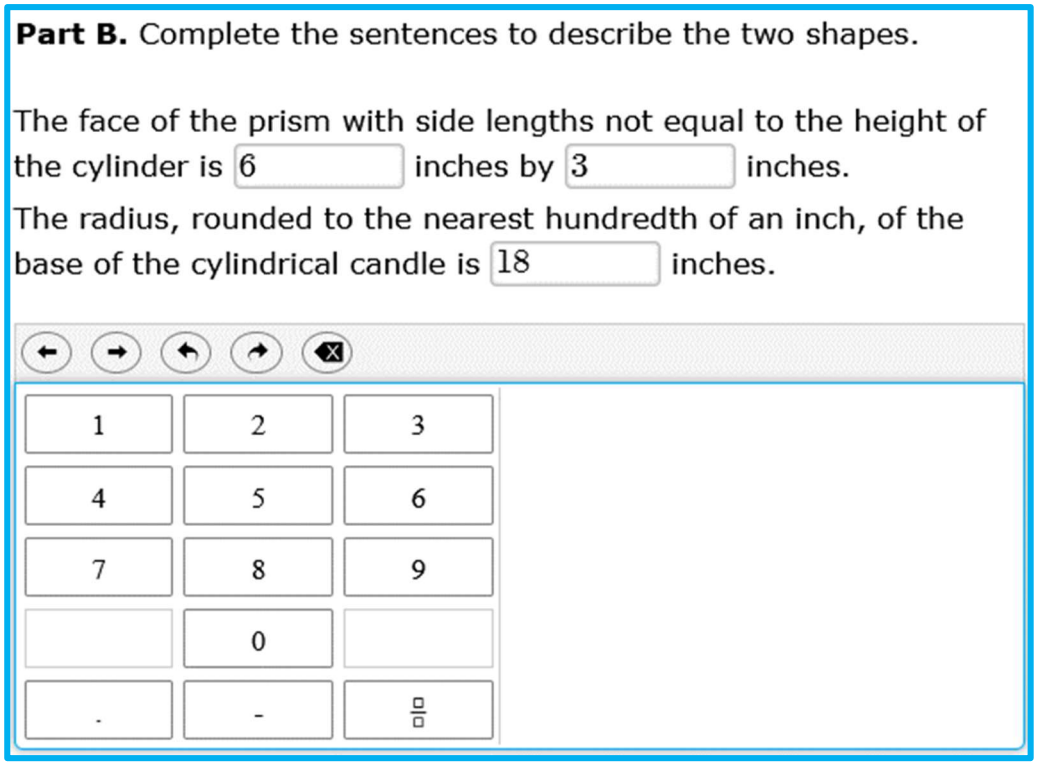
The figure shows the dimensions, in inches (in.), of a wax candle.



The candle is melted and remolded in the shape of a right cylinder of height 4 inches so that the volume remains the same.

Part A. Complete the sentence to relate the dimensions of the two shapes.

As the volumes of the two shapes are the same and the height of the cylinder is equal to one of the side lengths of the prism, the [circumference of the base] of the cylindrical candle must be equal to the [area] of the face of the prism with side lengths that are not equal to the height of the cylinder.



Notes on Scoring

This response earns no credit (0 points) because it shows incorrectly completed statements.

The student may confuse the area of a circle with the circumference of a circle but recognizes that since the height of the prism and the cylinder are the same, it must be equivalent to the area of the face of the prism.

The student then understands that since the height of the cylinder is 4 inches and the prism and cylinder have the same height, the side lengths of the face of the prism mentioned in part A are 6 inches and 3 inches. The student substitutes these values into the area formula to calculate the area of the face of the prism to get 18 inches. The student then incorrectly uses this value as the length of the radius.

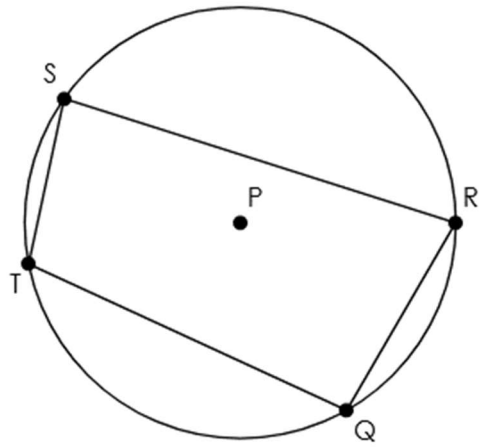
Geometry
Spring 2024 Item Release

Question 18

Question and Scoring Guidelines

Question 18

Quadrilateral QRST is inscribed in circle P.



Which equation must be true?

Ⓐ $m\angle Q = 90°$

Ⓑ $m\angle Q = m\angle S$

Ⓒ $m\angle Q + m\angle R = 180°$

Ⓓ $m\angle Q + m\angle S = 180°$

Points Possible: 1

Content Cluster: Understand and apply theorems about circles.

Content Standard: Construct the inscribed and circumscribed circles of a triangle; prove and apply the property that opposite angles are supplementary for a quadrilateral inscribed in a circle. *(G.C.3)*

Depth of Knowledge: Level 1

Modeling and Reasoning: No

Scoring Guidelines

Rationale for Option A: This is incorrect. The student may assume $\overline{TR}$ is a diameter of circle P and applies the theorem that angles inscribed in a semicircle are right.

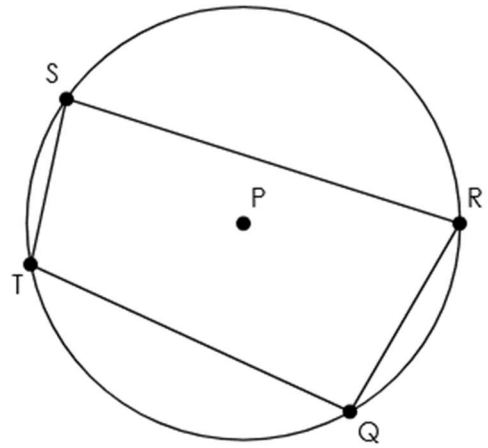
Rationale for Option B: This is incorrect. The student may assume PQRS is a parallelogram since it is a quadrilateral and applies the theorem that opposite angles of a parallelogram are equal in measure.

Rationale for Option C: This is incorrect. The student may confuse the opposite and adjacent angles in a quadrilateral, applying the theorem that opposite angles of a quadrilateral inscribed in a circle are supplementary because the arcs they intercept share endpoints but do not overlap to adjacent angles Q and R.

Rationale for Option D: **Key** – The student applies the theorem that opposite angles of a quadrilateral inscribed in a circle are supplementary because the arcs they intercept share endpoints but do not overlap. Therefore, the sum of opposite angles Q and S is 180°.

Sample Response: 1 point

Quadrilateral QRST is inscribed in circle P.



Which equation must be true?

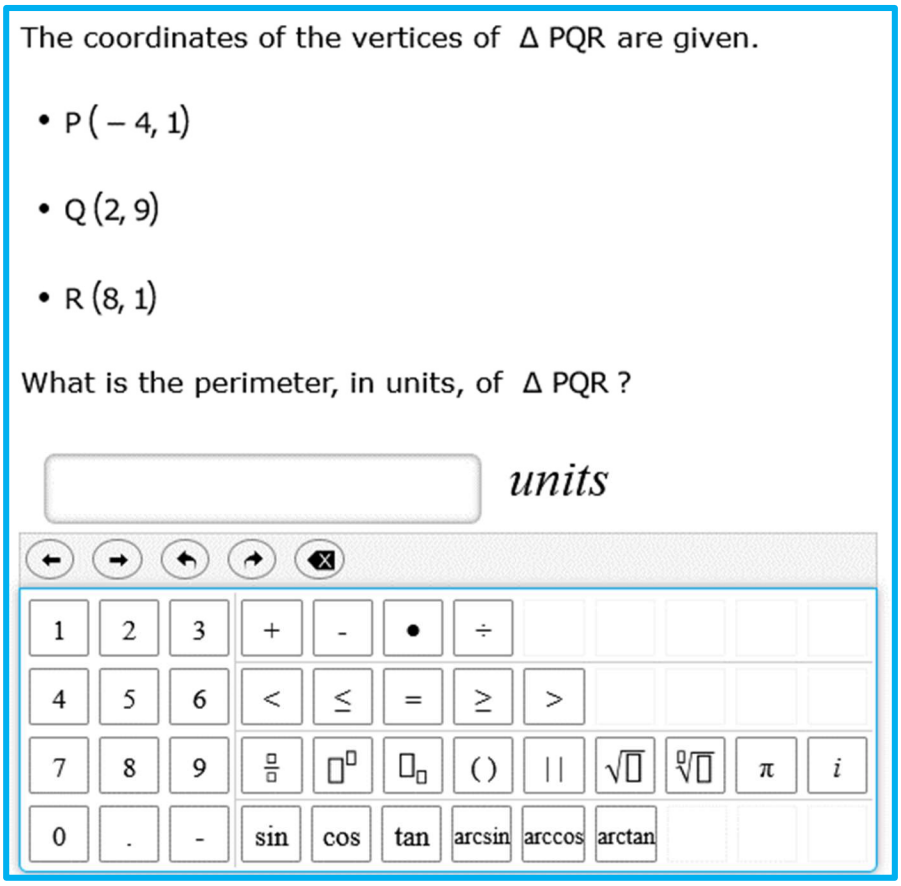
- Ⓐ $m\angle Q = 90°$
- Ⓑ $m\angle Q = m\angle S$
- Ⓒ $m\angle Q + m\angle R = 180°$
- ● $m\angle Q + m\angle S = 180°$

Geometry
Spring 2024 Item Release

Question 19

Question and Scoring Guidelines

Question 19



Points Possible: 1

Content Cluster: Use coordinates to prove simple geometric theorems algebraically and to verify specific geometric statements.

Content Standard: Use coordinates to compute perimeters of polygons and areas of triangles and rectangles, e.g., using the distance formula. *(G.GPE.7)*

Depth of Knowledge: Level 2

Modeling and Reasoning: No

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student correctly identifies the value, providing evidence of the ability to use coordinates to compute perimeters of polygons.

Exemplar Response

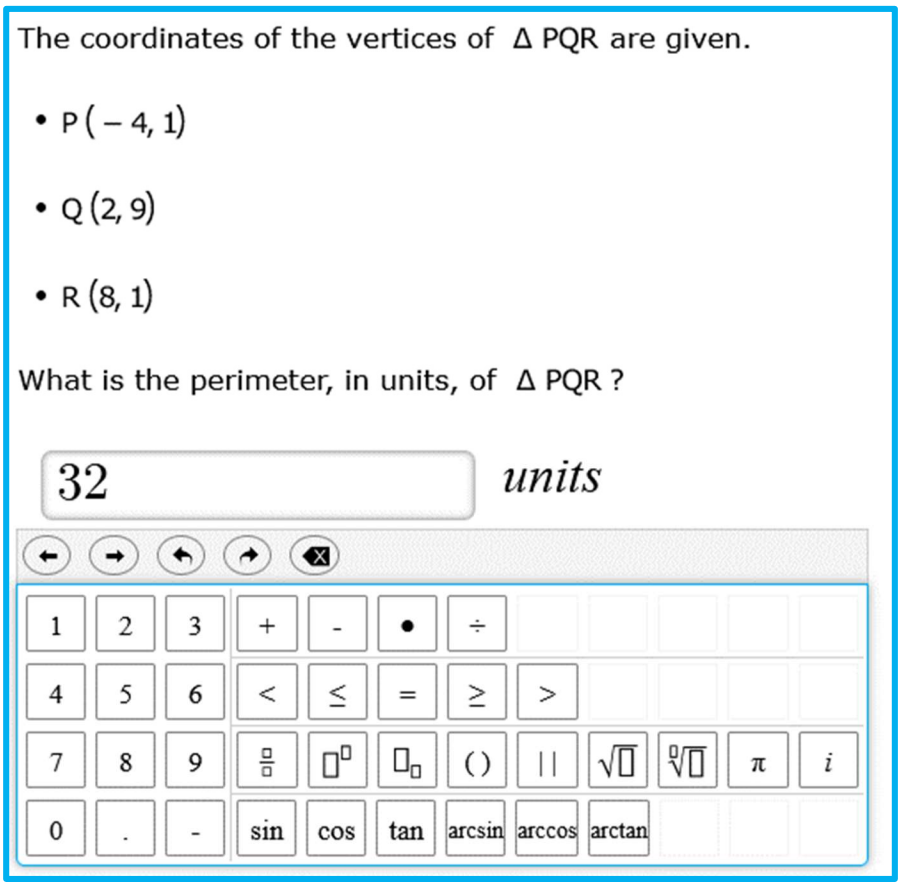
- 32 units
  or
- any equivalent value

Geometry
Spring 2024 Item Release

Question 19

Sample Responses

Sample Response: 1 point



Notes on Scoring

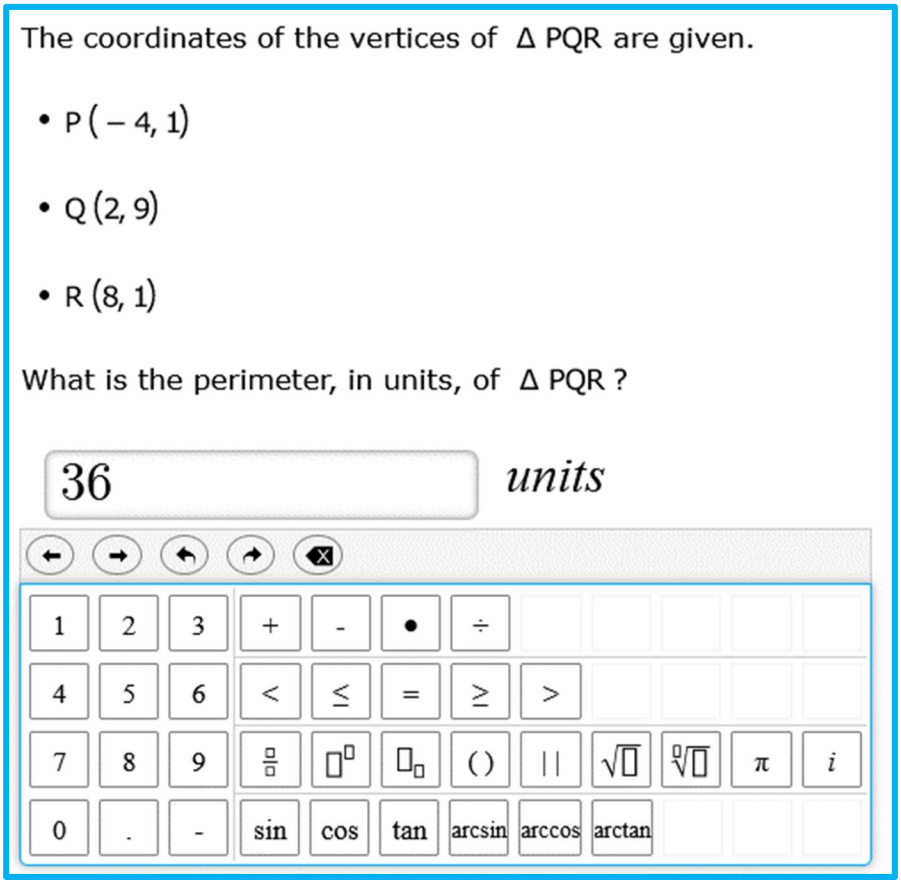
This response earns full credit (1 point) because it shows the correct perimeter of $\triangle PQR$.

The student uses the distance formula to calculate the length of the 3 sides of triangle PQR. The calculation for the length of side $\overline{PQ}$ is shown.

$$\begin{aligned}\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} &= \sqrt{(2 - (-4))^2 + (9 - 1)^2} \\ &= \sqrt{(2 + 4)^2 + (9 - 1)^2} \\ &= \sqrt{6^2 + 8^2} \\ &= \sqrt{36 + 64} \\ &= \sqrt{100} \\ &= 10\end{aligned}$$

Similarly, the distance formula is used to find the length of side $\overline{QR}$, $\sqrt{(8-2)^2 + (1-9)^2} = 10$, and the length of side $\overline{PR}$, $\sqrt{(8-(-4)^2 + (1-1)^2} = 12$. After determining the length of all three sides of the triangle, the three lengths are added together to calculate the perimeter of triangle PQR as 32 units.

Sample Response: 0 points



Notes on Scoring

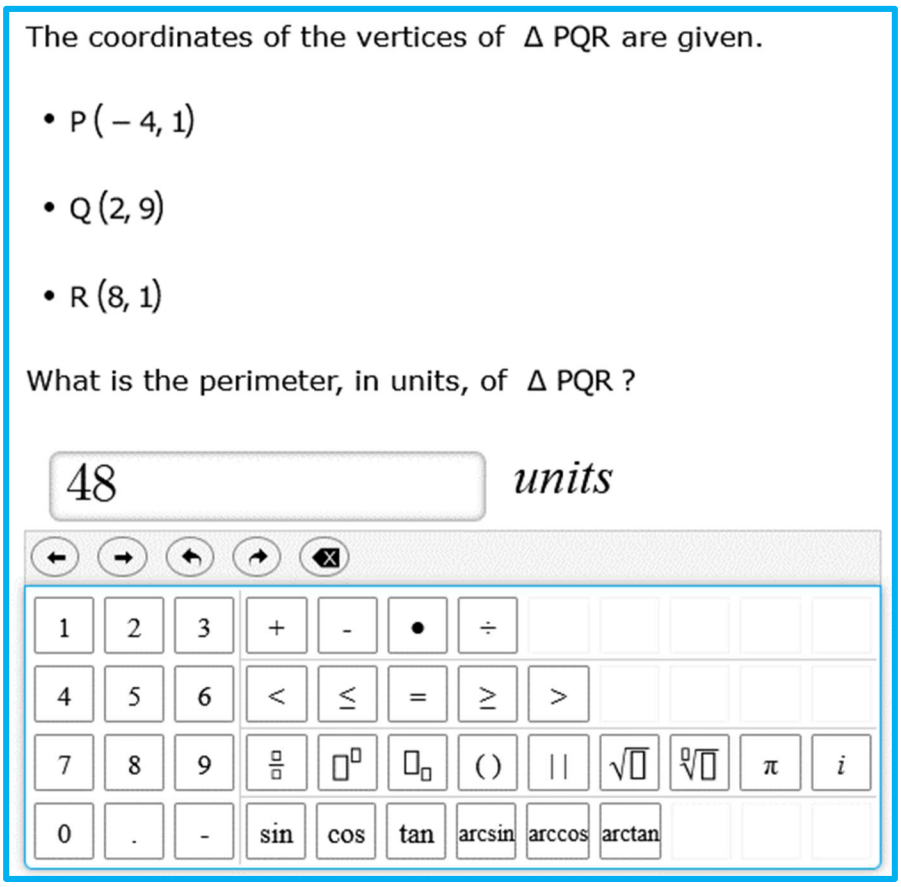
This response earns no credit (0 points) because it shows an incorrect perimeter of $\triangle PQR$.

The student may recognize that vertices P and R are collinear, and thus calculates the length of side $\overline{PR}$ as the absolute value of the difference of their x-coordinates.

$$|8-(-4)| = |8+4| \\ = 12$$

However, the student may think that all three sides are the same length and multiply 12 by 3 and state that the perimeter of triangle PQR is 36 units.

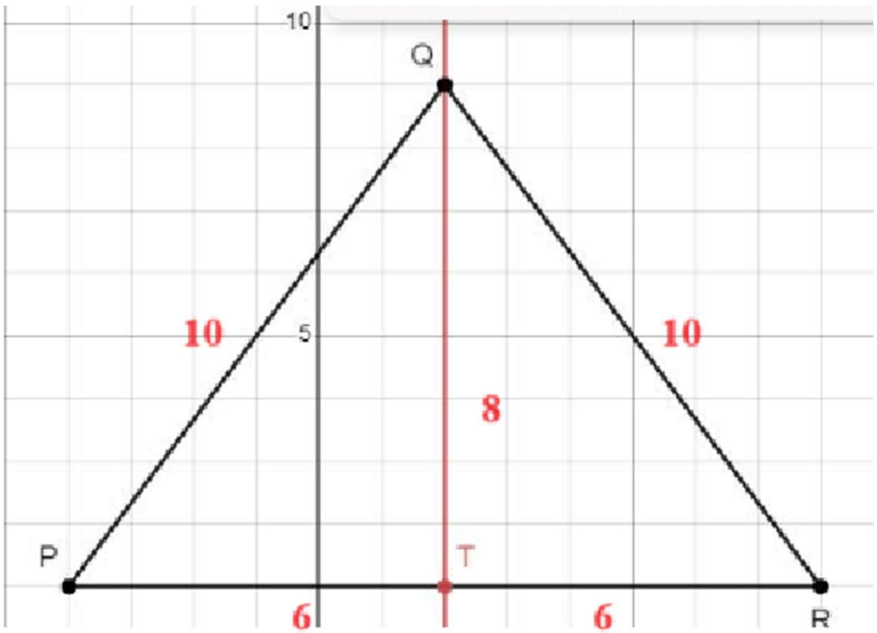
Sample Response: 0 points



Notes on Scoring

This response earns no credit (0 points) because it shows an incorrect perimeter of $\triangle PQR$.

The student may divide triangle PQR into two right triangles each having side lengths of 6, 8, and 10, as shown.



The student may then calculate the perimeter of the two triangles 2(6 + 8 + 10) = 48, instead of the perimeter of the PQR, 6 + 6 + 10 + 10 = 32.

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Question 20

Question and Scoring Guidelines

Question 20

The area of triangle QRS is 24 square centimeters. Triangle QRS is scaled by a factor of 4 to create triangle TUV.

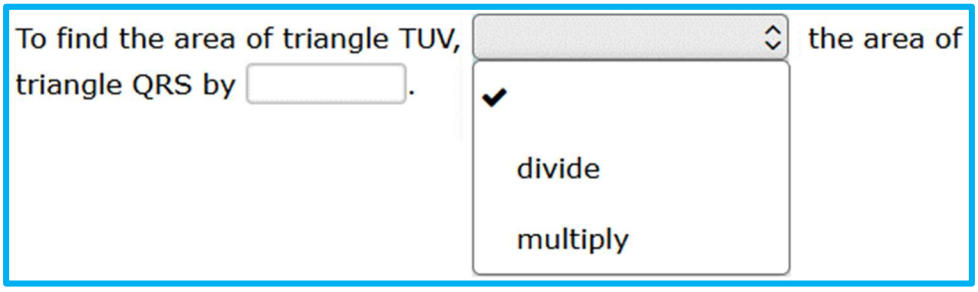
Complete the statement to describe a way to find the area of triangle TUV.

To find the area of triangle TUV, [ ] the area of triangle QRS by [ ].

The area of triangle TUV is [ ] square centimeters.



Drop Down Choices



Points Possible: 2

Content Cluster: Understand the relationships between lengths, area, and volumes.

Content Standard: When figures are similar, understand and apply the fact that when a figure is scaled by a factor of k, the effect on lengths, areas, and volumes is that they are multiplied by k, k^2, and k^3, respectively. *(G.GMD.6)*

Depth of Knowledge: Level 2

Modeling and Reasoning: No

Scoring Guidelines

For full credit (2 points), the student's response satisfies both bullets below.

For partial credit (1 point), the student's response satisfies one of the bullets below.

- The student correctly completes the first statement, providing evidence of the ability to understand the fact that when a figure is scaled by a factor of k, the effect on the area is that it is multiplied by k^2.

- The student calculates the area as 384, or a correct area based on an incorrect answer in the first sentence, providing evidence of the ability to apply the fact that when a figure is scaled by a factor of k, the effect on the area is that it is multiplied by k^2.

Exemplar Response

2 points

The area of triangle QRS is 24 square centimeters. Triangle QRS is scaled by a factor of 4 to create triangle TUV.

Complete the statement to describe a way to find the area of triangle TUV.

To find the area of triangle TUV, [multiply] the area of triangle QRS by [16].

The area of triangle TUV is [384] square centimeters.

or

The area of triangle QRS is 24 square centimeters. Triangle QRS is scaled by a factor of 4 to create triangle TUV.

Complete the statement to describe a way to find the area of triangle TUV.

To find the area of triangle TUV, [divide] the area of triangle QRS by [$\frac{1}{16}$].

The area of triangle TUV is [384] square centimeters.

1 point

The area of triangle QRS is 24 square centimeters. Triangle QRS is scaled by a factor of 4 to create triangle TUV.

Complete the statement to describe a way to find the area of triangle TUV.

To find the area of triangle TUV, [multiply] the area of triangle QRS by [4].

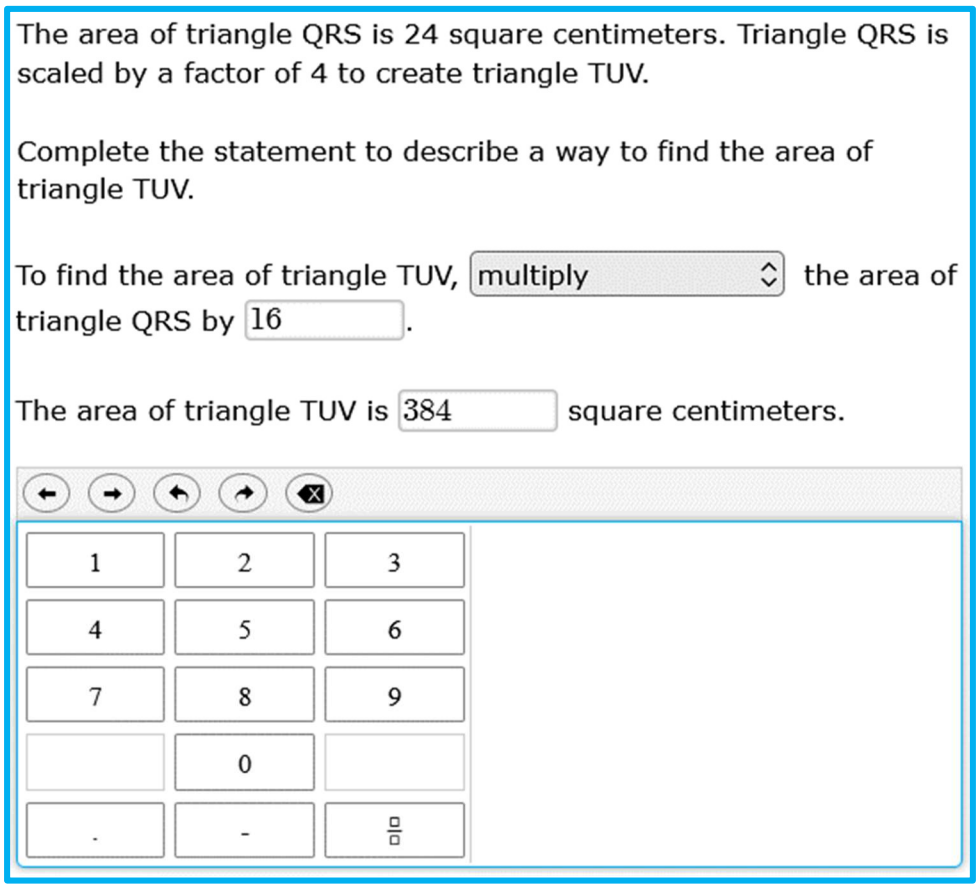
The area of triangle TUV is [96] square centimeters.

Geometry
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Question 20

Sample Responses

Sample Response: 2 points



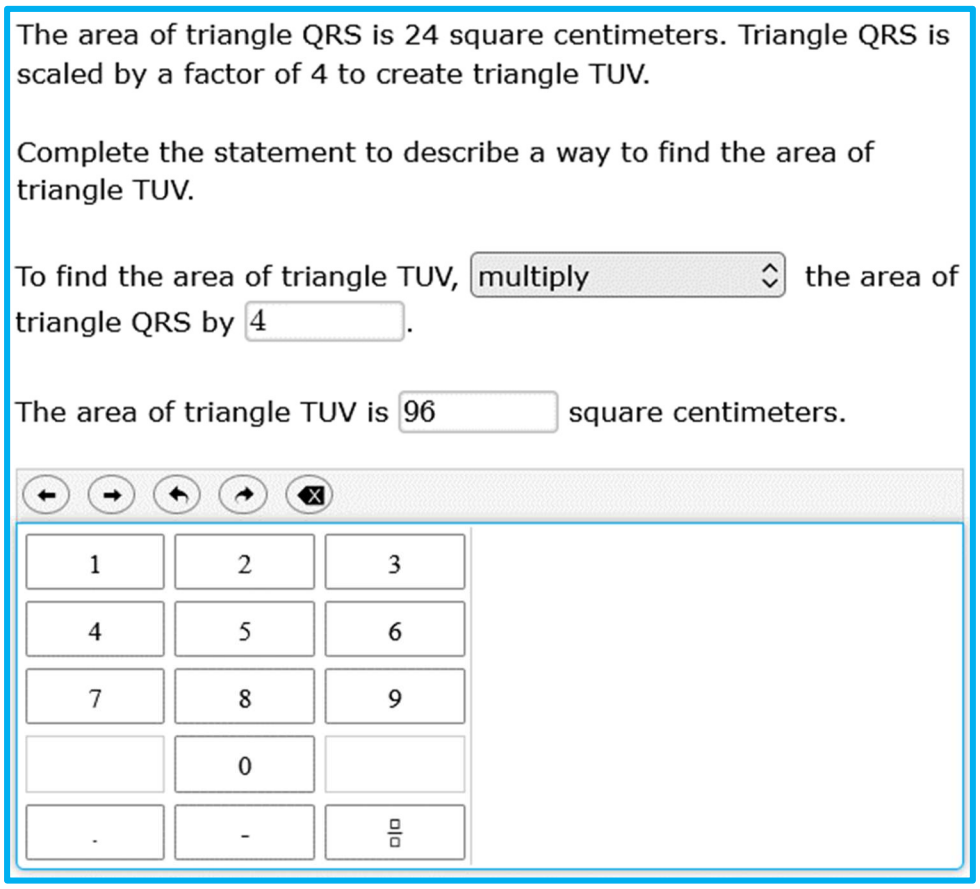
Notes on Scoring

This response earns full credit (2 points) because it correctly completes both statements.

The student may understand that since triangle QRS is scaled by a factor greater than 1 it will be increasing in size, signifying multiplication will have to be used. Since all sides of triangle QRS are multiplied by a factor of 4 to create triangle TUV and the area of a triangle is half of the triangle's height multiplied by the length of its base, the scale factor must be squared, $4^2 = 16$, and multiplied by the area of triangle QRS to calculate the area of triangle TUV.

Performing this multiplication, 24×16, gives the area of TUV as 384 square centimeters.

Sample Response: 1 point

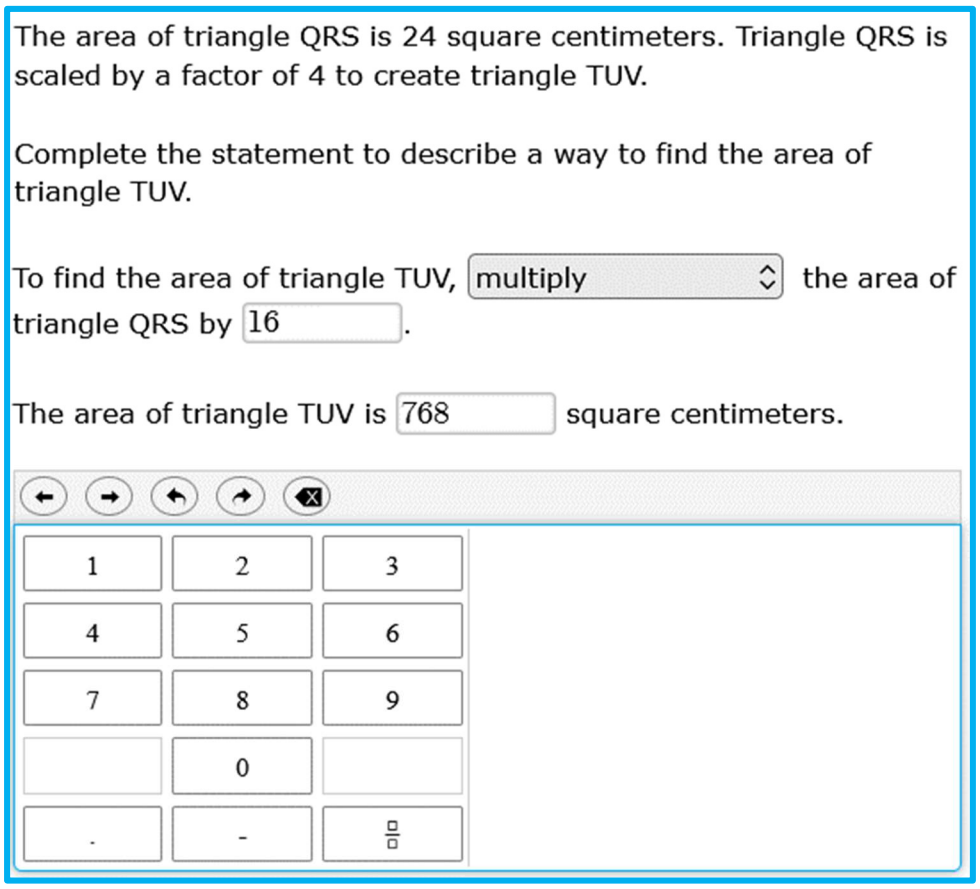


Notes on Scoring

This response earns partial credit (1 point) because it shows an area based on the wrong multiplier.

The student may understand that since triangle QRS is scaled by a factor greater than 1 it will be increasing in size, signifying multiplication will have to be used. However, the student does not consider the fact that area is a two-dimensional measurement and multiplies the original area by 4 instead of 4^2. Thus, the student thinks the area of TUV is 96 square centimeters.

Sample Response: 1 point

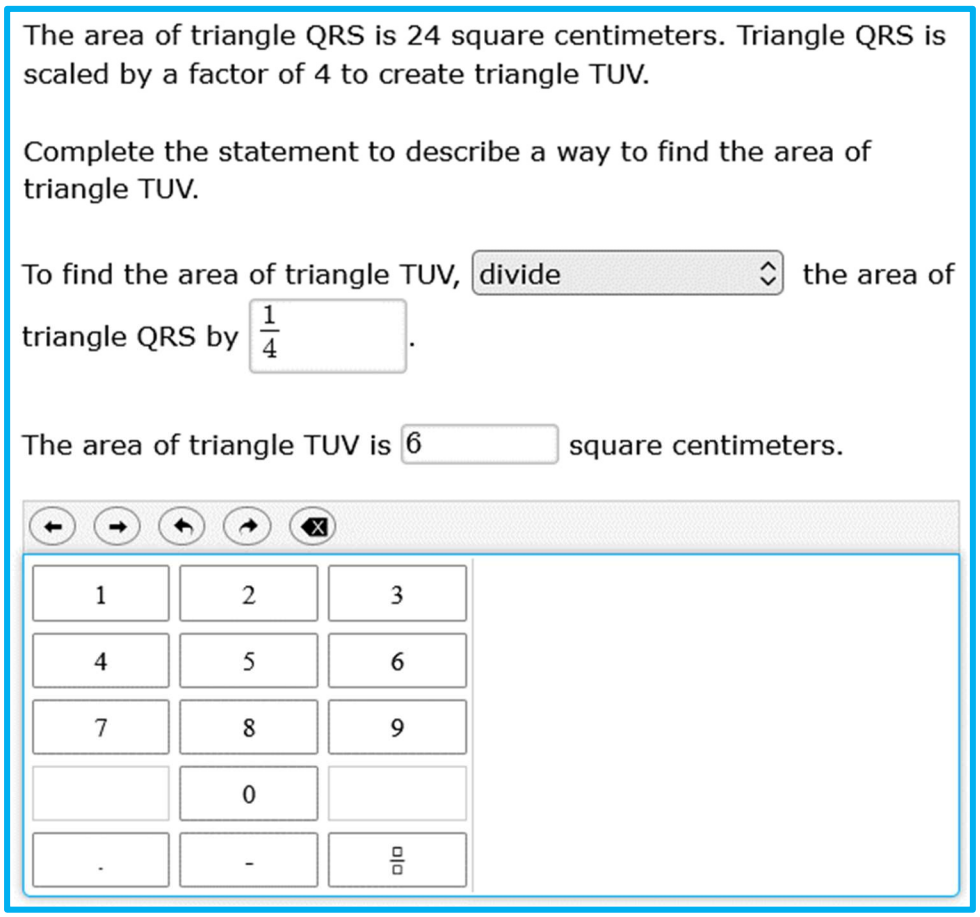


Notes on Scoring

This response earns partial credit (1 point) because it shows an area based on the wrong multiplier.

The student may understand that multiplication is used to create triangle TUV since the scale factor is greater than 1, and that in order to calculate the area of triangle TUV, the area of QRS can be multiplied by the square of the scale factor. However, the student may not realize that 24 describes the area of QRS, and instead thinks it is one of the dimensions of the triangle along with 4. As a result, the student calculates the area of QRS using the formula for the area of a triangle, $\frac{1}{2}(24)(4) = 48$ square centimeters. Next, the student multiplies 48 by 16 to calculate the area of TUV as 768 square centimeters.

Sample Response: 0 points

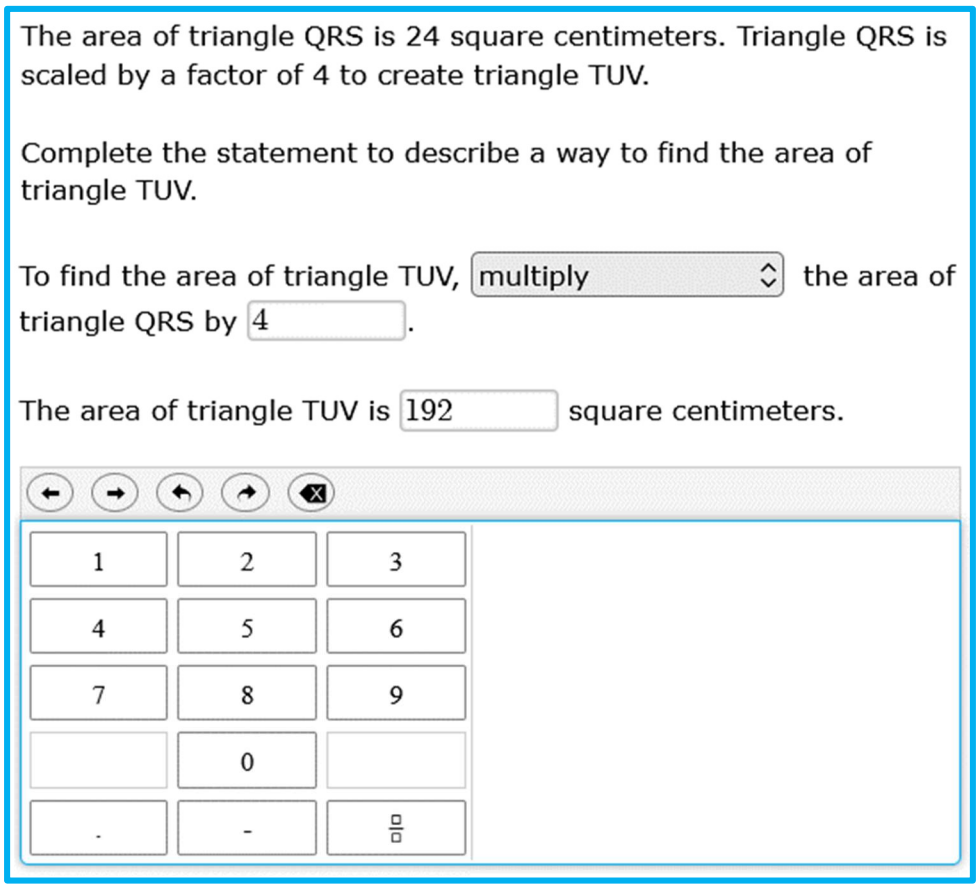


Notes on Scoring

This response earns no credit (0 points) because it incorrectly completes the statements.

The student may understand that dividing by $\frac{1}{4}$ is the same as multiplying by 4 but uses the scale factor k for side length instead of k^2 for area.

Sample Response: 0 points



Notes on Scoring

This response earns no credit (0 points) because it incorrectly completes the statements.

While the student may understand multiplication should be used since triangle QRS is scaled by a factor greater than 1 and thus increasing in size, the student may not consider the fact that the area of a triangle is half of the length of its base times its height and multiplies the area of triangle QRS by 4 even though both the base and height of triangle QRS are increased by a factor of 4, meaning the area of triangle QRS should be multiplied by a factor of 4^2 instead of 4.

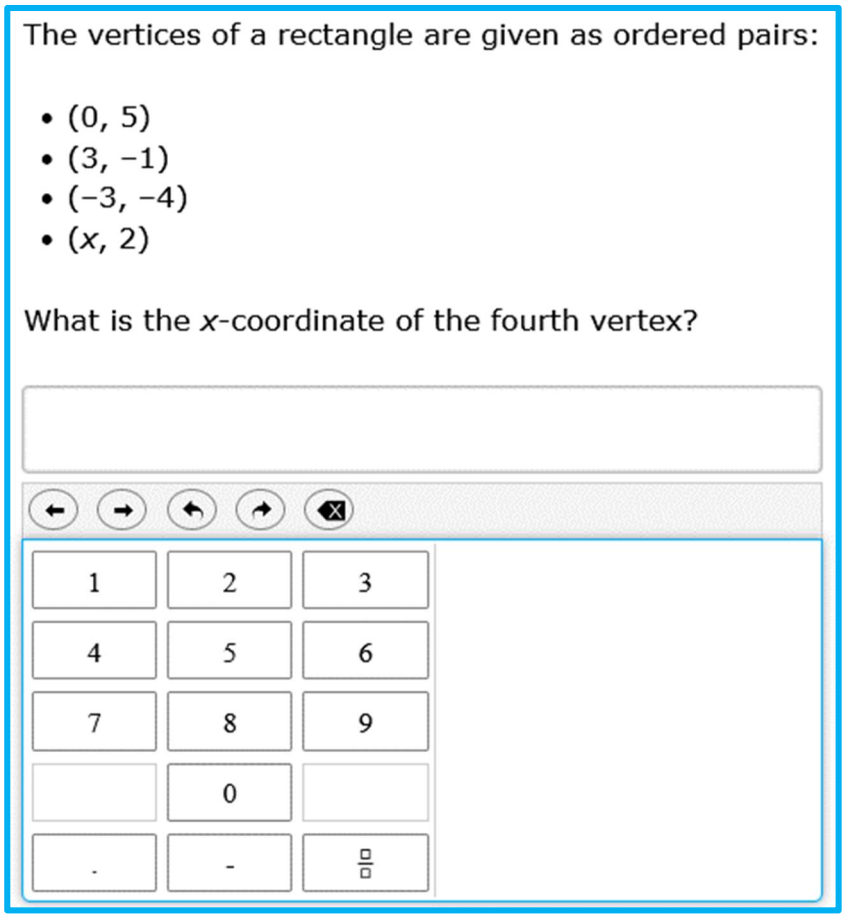
Additionally, the student may consider 24 as a side length of triangle QRS instead of the area of triangle QRS, with the scale factor of 4 as the other dimension of triangle QRS. The student then calculates the area of triangle QRS as $\frac{1}{2}(24)(4) = 48$, and then multiplies this value by the scale factor of 4 to find the area of TUV to be 192 square centimeters.

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Question 21

Question and Scoring Guidelines

Question 21



Points Possible: 1

Content Cluster: Use coordinates to prove simple geometric theorems algebraically and to verify specific geometric statements.

Content Standard: Use coordinates to prove simple geometric theorems algebraically and verify geometric relationships algebraically, including properties of special triangles, quadrilaterals, and circles. *For example, determine if a figure defined by four given points in the coordinate plane is a rectangle; determine if a specific point lies on a given circle. (G.GPE.4)*

Depth of Knowledge: Level 2

Modeling and Reasoning: Yes

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student identifies the correct coordinate, providing evidence of the ability to use coordinates to prove simple theorems algebraically.

Exemplar Response

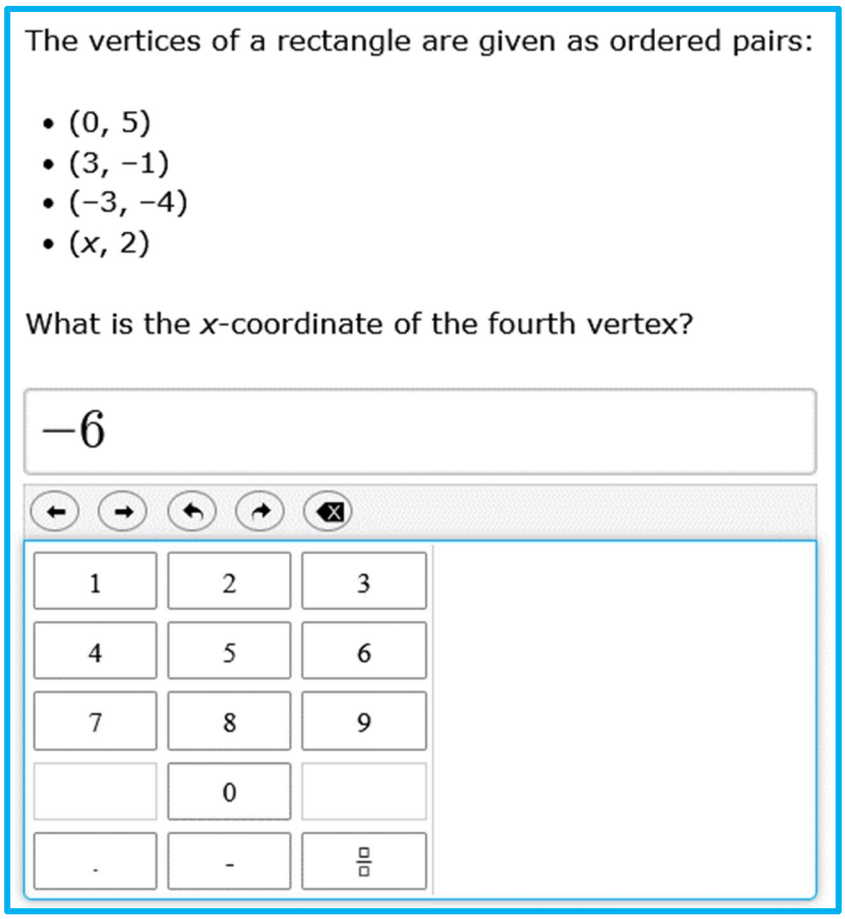
- –6
  or
- any equivalent value

Geometry
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Question 21

Sample Responses

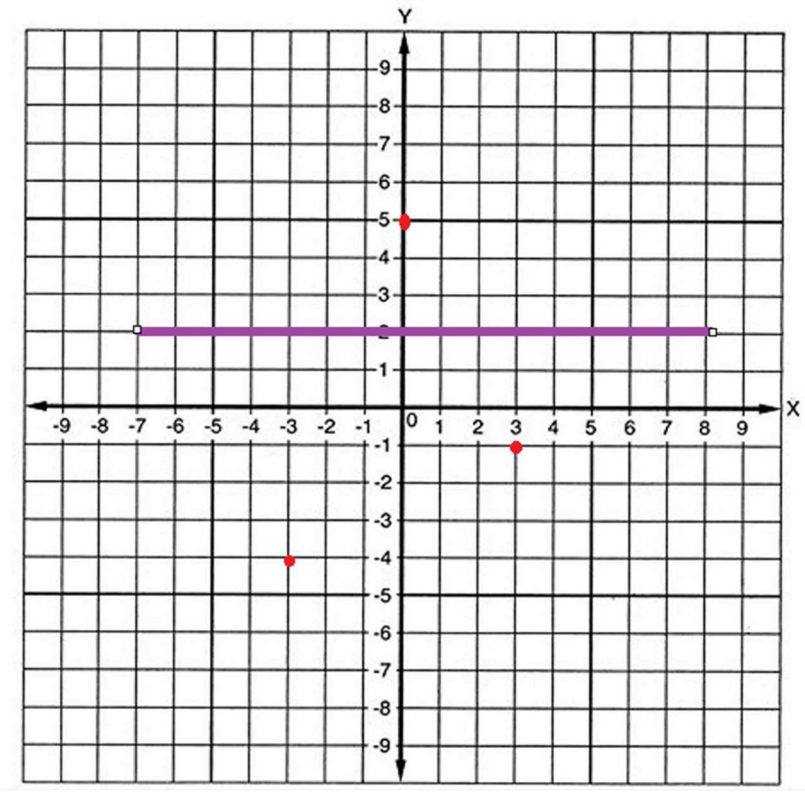
Sample Response: 1 point



Notes on Scoring

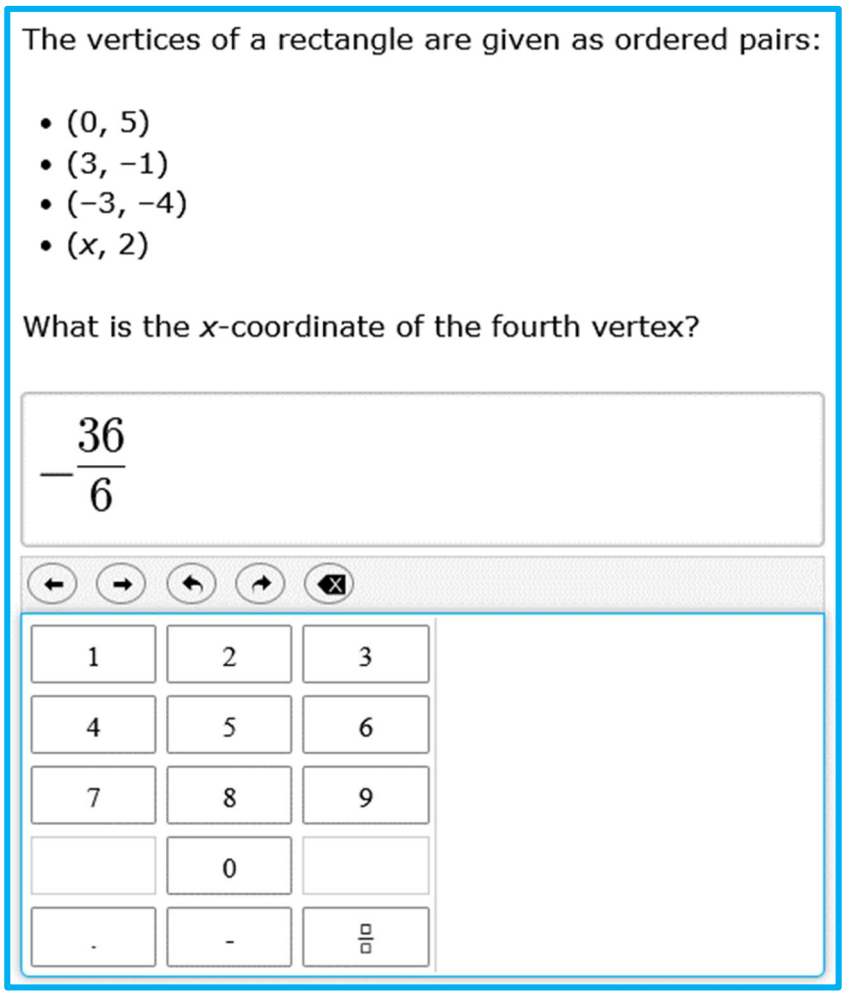
This response earns full credit (1 point) because it shows the correct value of the x-coordinate.

The student may sketch the three points given and recognize that the fourth point would have to be to the left of –3 for the points to make a rectangle since it must lie on the line $y = 2$.



To find the missing vertex, the student may use the fact that opposite sides of a rectangle are parallel, so they have the same slope to find the missing vertex. The student calculates the slope of the line going through (0, 5) and (3, –1) as $-\frac{6}{3}$, and counts up 6 and to the left 3 from the point (–3, –4), to find the point (–6, 2). Since this point is on the line $y = 2$, it is the fourth vertex of the rectangle. Thus, the x-coordinate of the fourth vertex is –6.

Sample Response: 1 point



Notes on Scoring

This response earns full credit (1 point) because it shows the correct value of the x-coordinate.

The student may understand the opposite sides of rectangles are parallel, which means the slope of the opposite sides must be equal. Plotting the three known ordered pairs, the student identifies the first two ordered pairs must be adjacent vertices of the rectangle in order for the last vertex to have the *y*-coordinate 2. The student then uses the slope formula and the first two vertices of the rectangle to find the slope of one of the sides of the rectangle.

$$\begin{aligned}\frac{\Delta y}{\Delta x} &= \frac{5-(-1)}{0-3} \\ &= \frac{5+1}{0-3} \\ &= -\frac{6}{3}\end{aligned}$$

Using the same formula with the second and third vertices, the student finds the slope of the adjacent side to be $\frac{3}{6}$, confirming the sides are perpendicular by definition. Next, the student sets the slope of the side formed with the first two vertices equal to the slope of the last two vertices since it appears they are opposite sides when graphed and solves the equation to determine the missing coordinate.

$$\begin{aligned}-\frac{6}{3} &= \frac{\Delta y}{\Delta x} \\ -\frac{6}{3} &= \frac{2-(-4)}{x-(-3)} \\ -\frac{6}{3} &= \frac{2+4}{x+3} \\ \frac{6}{-3} &= \frac{6}{x+3}\end{aligned}$$

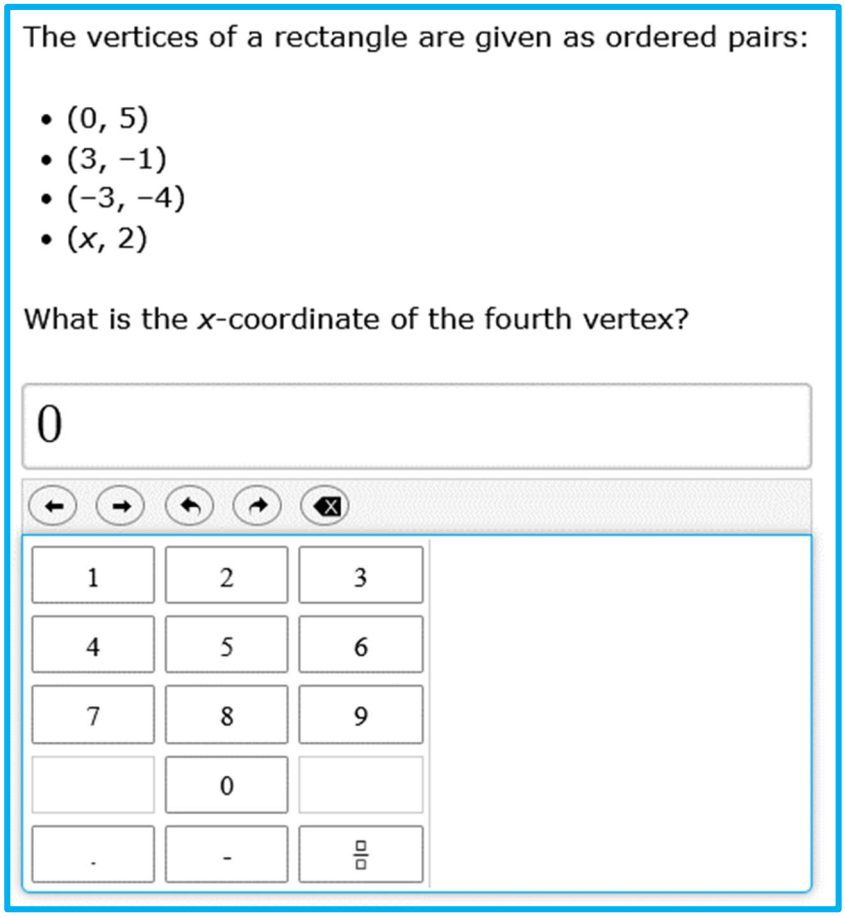
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(continued)

Notes on Scoring

Next, the student determines $-3(x+3)$ to be the common denominator of the fractions in the equation, and thus multiplies the fraction on the left side of the equals sign by $\frac{x+3}{x+3}$ and the fraction on the right side of the equal sign by $\frac{-3}{-3}$. This results in the new equation $\frac{6x+18}{-3x-9} = \frac{-18}{-3x-9}$. Since the fractions are equivalent, the student recognizes that the numerators of the fractions must be the same, thus $6x+18=-18$. First, the 18 is subtracted from both sides of this equation, resulting in $6x=-36$. Finally, the student divides both sides of the equation by 6 to conclude that the missing x-coordinate of the fourth ordered pair is $-\frac{36}{6}$.

Sample Response: 0 points

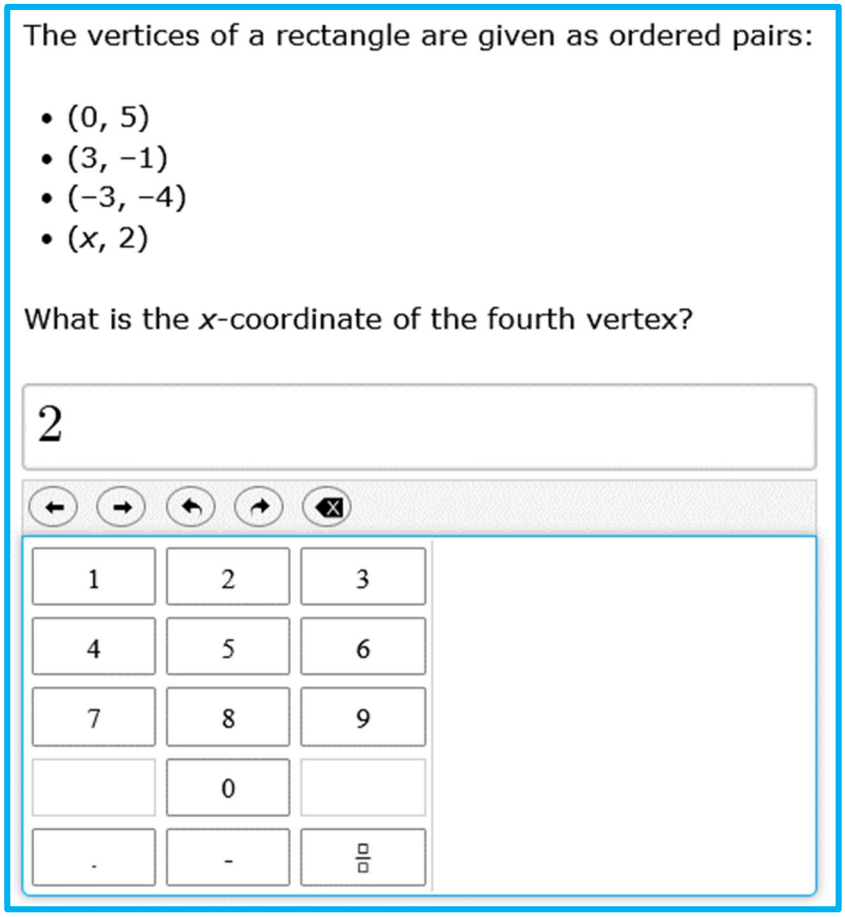


Notes on Scoring

This response earns no credit (0 points) because it shows an incorrect value of the x-coordinate.

The student may not have considered the negative in the x-value of the vertex (–3, –4), confusing it to be (3, –4). Understanding that the sides of the rectangles are parallel, the student believes that since there is no change in x-value between the second and third vertex listed, there should be no change in the x-values between the other two vertices either. Thus, the student finds the missing x-value to be 0.

Sample Response: 0 points



Notes on Scoring

This response earns no credit (0 points) because it shows an incorrect value of the x-coordinate.

The student may understand the change in x-coordinates and change in *y*-coordinates between the first two points in the list affects the missing coordinate since the first two points determine a side of the rectangle. However, the student may not consider the negative sign in the *y*-coordinate of the second vertex nor recognize this difference in coordinates is useful by establishing the slope of both parallel sides and instead notices 3, the change in the x-coordinates, is 1 less than 4, the change in *y*-coordinates. The student may then identify the change in the *y*-coordinates of the other pair of vertices as 6. Applying a possible pattern, the student thinks the change in the x-coordinates of the points must be 1 less than 6, or 5, and creates the equation shown to determine the missing coordinate.

$$\begin{aligned} x-(-3) &= 5 \\ x+3 &= 5 \\ x &= 5-3 \\ x &= 2 \end{aligned}$$

Therefore, the student thinks the missing coordinate is 2.

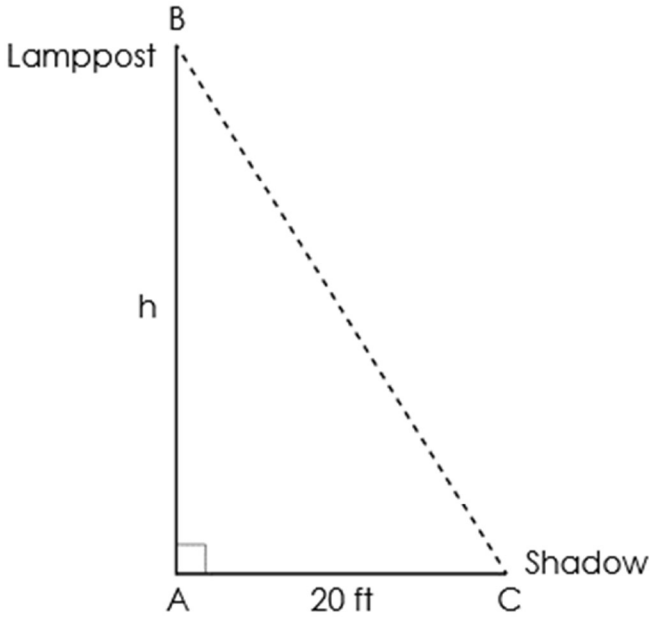
Geometry
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Question 22

Question and Scoring Guidelines

Question 22

A lamppost casts a shadow of 20 feet (ft) along the ground, as shown.



The measure of angle C is 58 degrees.

What is the height, *h*, of the lamppost to the nearest foot?

feet

Points Possible: 1

Content Cluster: Define trigonometric ratios and solve problems involving right triangles.

Content Standard: Solve problems involving right triangles.
a. Use trigonometric ratios and the Pythagorean Theorem to solve right triangles in applied problems if one of the two acute angles and a side length is given. *(G.SRT.8)*

Depth of Knowledge: Level 2

Modeling and Reasoning: No

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student identifies the correct height, providing evidence of the ability to use trigonometric ratios in applied problems.

Exemplar Response

- 32
  or
- any value between 32 and 32.01, inclusive

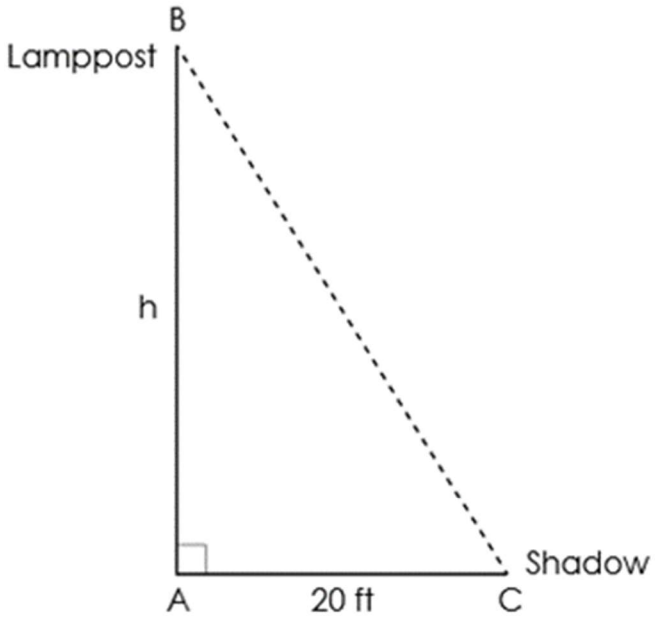
Geometry
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Question 22

Sample Responses

Sample Response: 1 point

A lamppost casts a shadow of 20 feet (ft) along the ground, as shown.



The measure of angle C is 58 degrees.

What is the height, *h*, of the lamppost to the nearest foot?

32 *feet*

Notes on Scoring

This response earns full credit (1 point) because it shows the correct height of the lamppost.

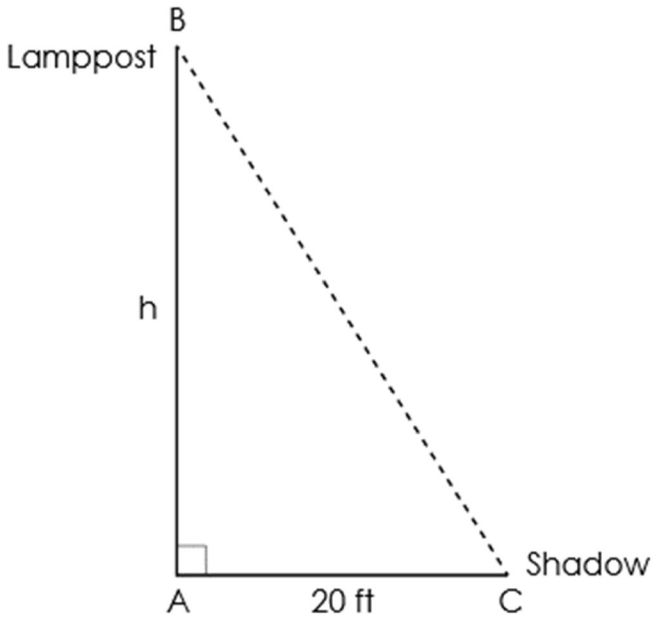
The student may realize that since the side adjacent to angle C is known and they need to find the length of the side opposite of angle C, they can use the tangent formula to find the value of *h*.

$$\tan(58°) = \frac{h}{20}$$
$$20\tan(58°) = h$$

After performing the calculation, the value of *h* is approximately 32 feet when rounded to the nearest foot.

Sample Response: 0 points

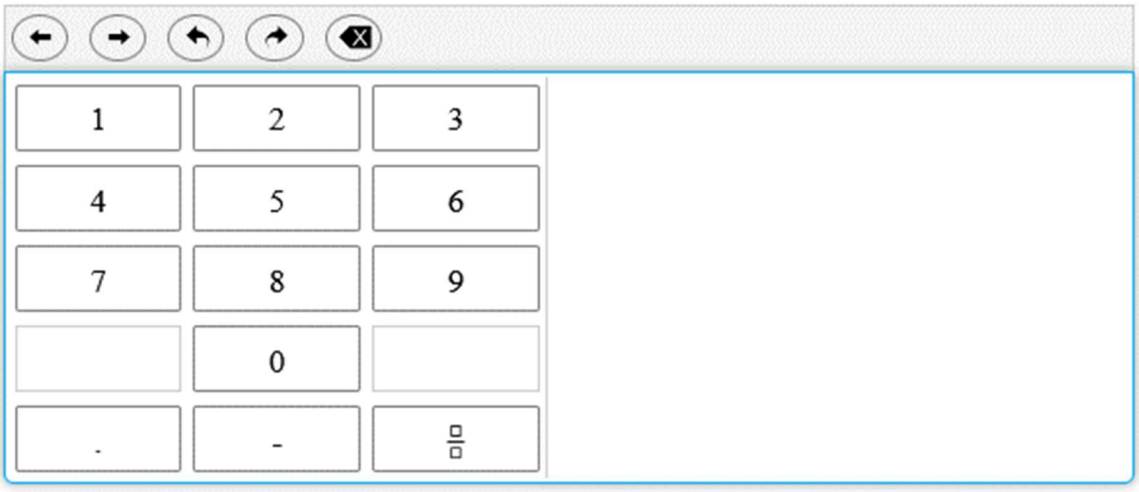
A lamppost casts a shadow of 20 feet (ft) along the ground, as shown.



The measure of angle C is 58 degrees.

What is the height, *h*, of the lamppost to the nearest foot?

38 *feet*



Notes on Scoring

This response earns no credit (0 points) because it shows an incorrect height of the lamppost.

The student may confuse the trigonometric formulas and think that the cosine formula should be used to find the height of the lamppost instead of the tangent formula and the cosine ratio to be the length of the adjacent side divided by the length of the opposite side instead of the length of the adjacent side divided by the length of the hypotenuse.

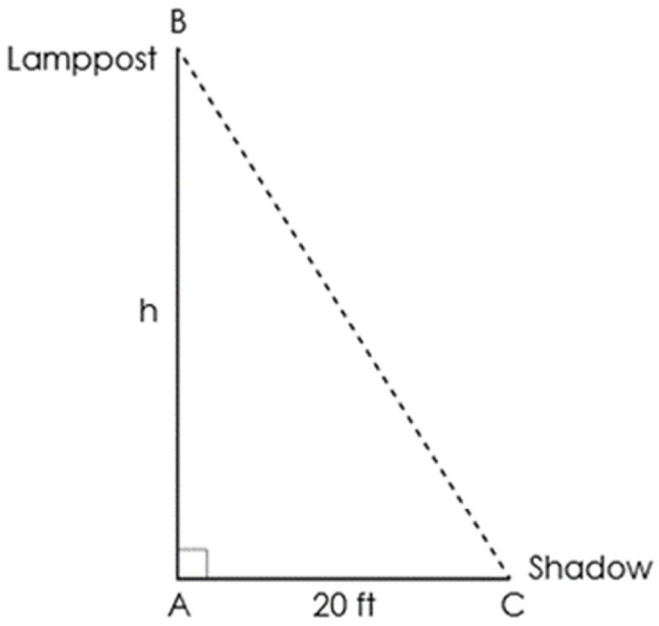
$$\cos(58°) = \frac{20}{h}$$

$$h = \frac{20}{\cos(58°)}$$

After performing the calculation, the value of *h* is approximately 38 feet when rounded to the nearest foot.

Sample Response: 0 points

A lamppost casts a shadow of 20 feet (ft) along the ground, as shown.



The measure of angle C is 58 degrees.

What is the height, *h*, of the lamppost to the nearest foot?

1160 *feet*

Notes on Scoring

This response earns no credit (0 points) because it shows an incorrect height of the lamppost.

The student may correctly identify that, given the information in the question, the tangent formula should be used to find the height of the lamppost, but incorrectly writes the left side of the equation as 58 instead of tan(58°).

$$\begin{aligned} 58 &= \frac{h}{20} \\ 20(58) &= h \\ 1160 &= h \end{aligned}$$

Thus, the student calculates the height of the lamppost as 1160 feet.

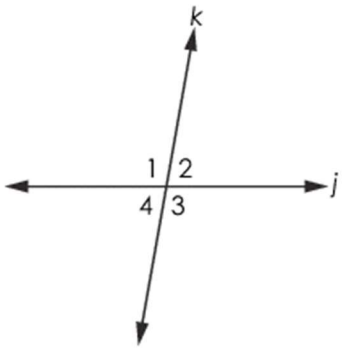
Geometry
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Question 23

Question and Scoring Guidelines

Question 23

Lines j and k intersect to form angles 1, 2, 3, and 4, as shown.



Eliot is trying to complete the proof that angle 3 is congruent to angle 1. First, he notes that angles 1 and 2 form a linear pair, as do angles 2 and 3. Next, he notes that the measures of angles 1 and 2 have a sum of 180 degrees because they are supplementary. Next, he notes that the measures of angles 2 and 3 have a sum of 180 degrees because they are supplementary.

Which statement should be the next step in Eliot's proof?

Ⓐ Angles 1 and 4 form a linear pair.

Ⓑ Angles 1 and 2 each have a measure of 90 degrees.

Ⓒ The measure of angle 1 and the measure of angle 3 have a sum of 180 degrees by substitution.

Ⓓ The sum of the measures of angles 1 and 2 is equal to the sum of the measures of angles 2 and 3 by substitution.

Points Possible: 1

Content Cluster: Prove geometric theorems both formally and informally using a variety of methods.

Content Standard: Prove and apply theorems about lines and angles. Theorems include but are not restricted to the following: vertical angles are congruent; when a transversal crosses parallel lines, alternate interior angles are congruent and corresponding angles are congruent; points on a perpendicular bisector of a line segment are exactly those equidistant from the segment's endpoints. *(G.CO.9)*

Depth of Knowledge: Level 3

Modeling and Reasoning: Yes

Scoring Guidelines

Rationale for Option A: This is incorrect. The student may identify a true statement, but one that does not further the proof.

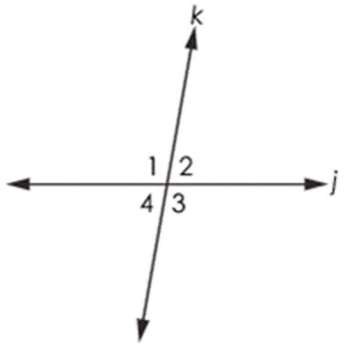
Rationale for Option B: This is incorrect. The student may assume all linear pairs are created by two 90° angles.

Rationale for Option C: This is incorrect. The student may think supplementary angles are transitive, and so angles 1 and 3 are supplementary because angle 1 is supplementary to angle 2 and angle 2 is supplementary to angle 3.

Rationale for Option D: **Key** – The student notes that two pairs of angle measures with the same sum would be equal to each other by substitution.

Sample Response: 1 point

Lines j and k intersect to form angles 1, 2, 3, and 4, as shown.



Eliot is trying to complete the proof that angle 3 is congruent to angle 1. First, he notes that angles 1 and 2 form a linear pair, as do angles 2 and 3. Next, he notes that the measures of angles 1 and 2 have a sum of 180 degrees because they are supplementary. Next, he notes that the measures of angles 2 and 3 have a sum of 180 degrees because they are supplementary.

Which statement should be the next step in Eliot's proof?

Ⓐ Angles 1 and 4 form a linear pair.

Ⓑ Angles 1 and 2 each have a measure of 90 degrees.

Ⓒ The measure of angle 1 and the measure of angle 3 have a sum of 180 degrees by substitution.

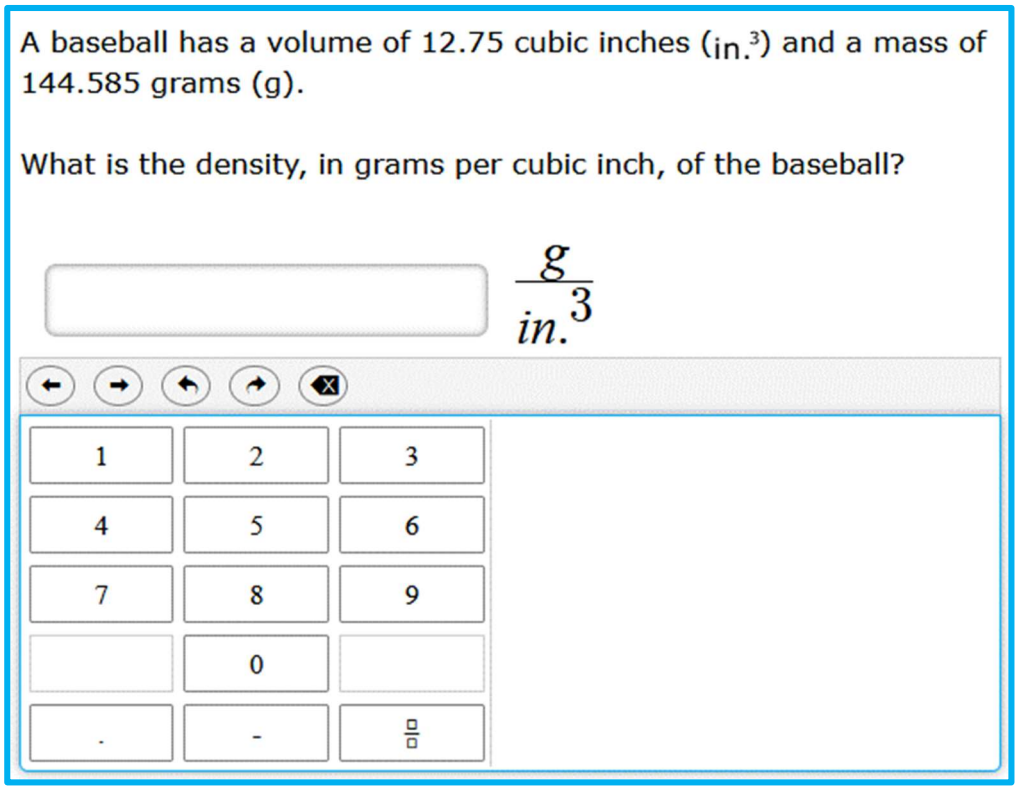
● The sum of the measures of angles 1 and 2 is equal to the sum of the measures of angles 2 and 3 by substitution.

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Question 26

Question and Scoring Guidelines

Question 26



Points Possible: 1

Content Cluster: Apply geometric concepts in modeling situations.

Content Standard: Apply concepts of density based on area and volume in modeling situations (*e.g., persons per square mile, BTUs per cubic foot*). *(G.MG.2)*

Depth of Knowledge: Level 1

Modeling and Reasoning: No

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student correctly finds the density, providing evidence of the ability to calculate the density of an object.

Exemplar Response

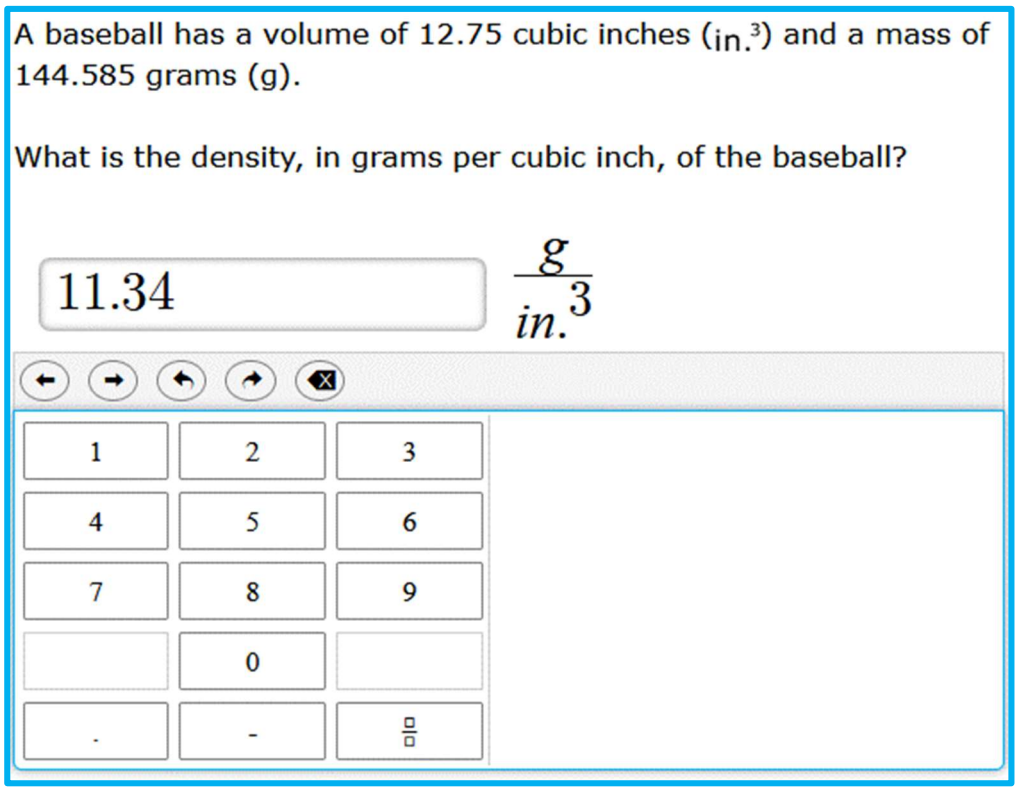
- $11.34 \frac{\text{g}}{\text{in.}^3}$
  or
- any equivalent value

Geometry
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Question 26

Sample Responses

Sample Response: 1 point

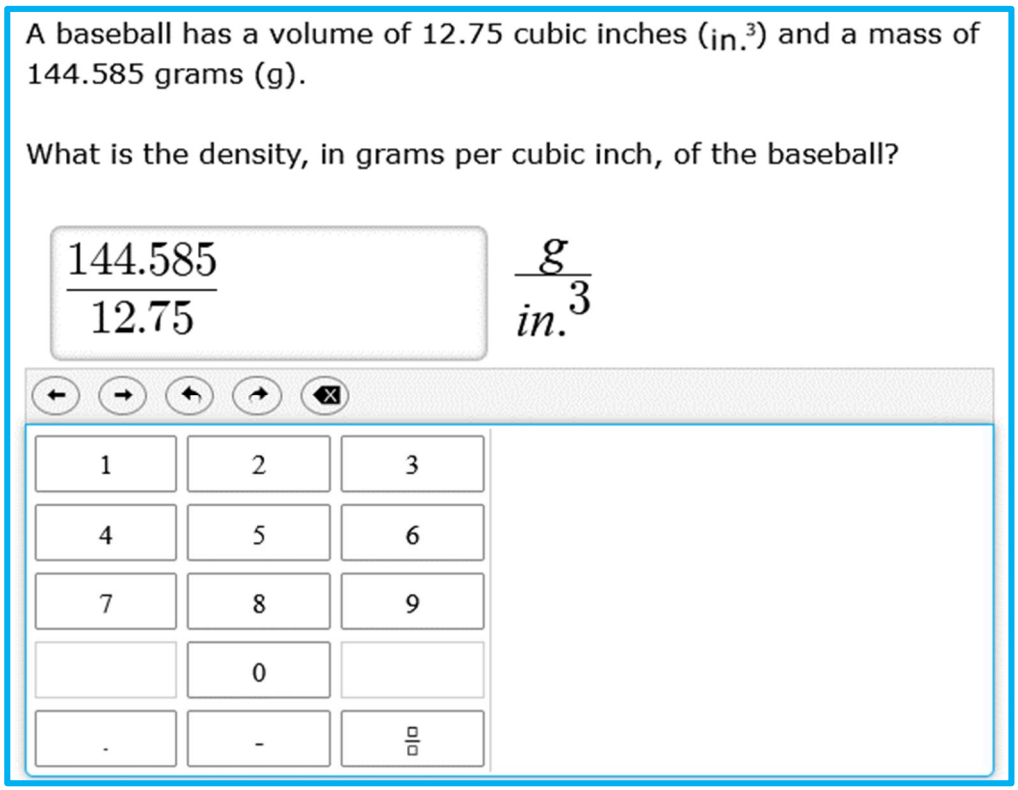


Notes on Scoring

This response earns full credit (1 point) because it shows a correct value for the density of the baseball.

The student may recall that density of an object can be determined by dividing the total mass of the object by its volume: $\frac{144.585}{12.75} = 11.34$. Thus, the density of the baseball is 11.34 $\frac{g}{in.^3}$.

Sample Response: 1 point

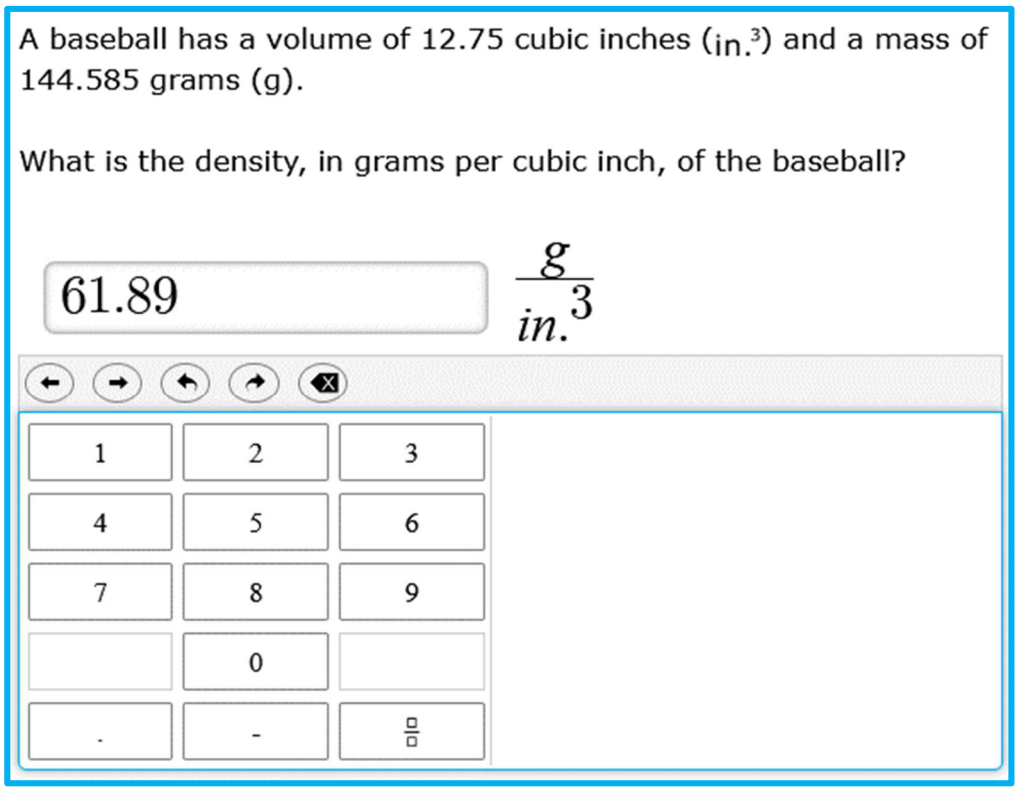


Notes on Scoring

This response earns full credit (1 point) because it shows a correct value for the density of the baseball.

The student may recall that density of an object can be determined by dividing the total mass of the object by its volume, writing this quotient as the fraction $\frac{144.585}{12.75}\frac{g}{in.^3}$.

Sample Response: 0 points

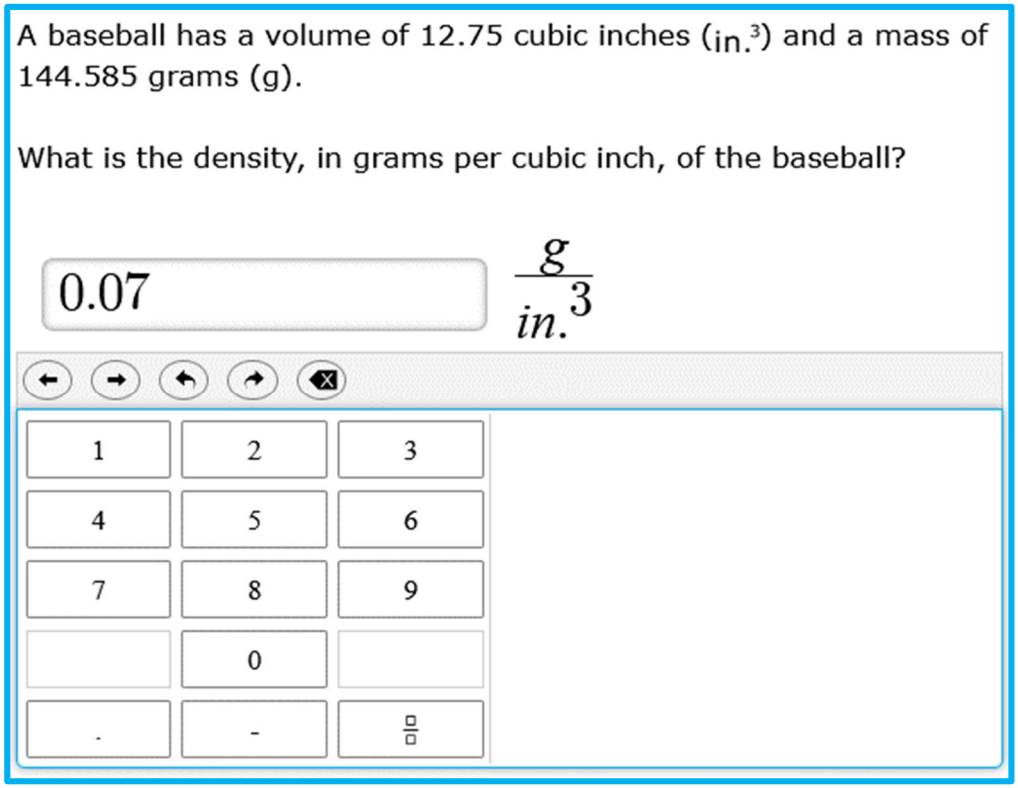


Notes on Scoring

This response earns no credit (0 points) because it shows an incorrect value for the density of the baseball.

The student may think that they need to take the cube root of the baseball's volume before applying the density formula so that no units have exponents: $\sqrt[3]{12.753} \approx 2.34$. The student then applies the density formula by dividing 144.585 by 2.34 to get a density of about 61.89.

Sample Response: 0 points



Notes on Scoring

This response earns no credit (0 points) because it shows an incorrect value for the density of the baseball.

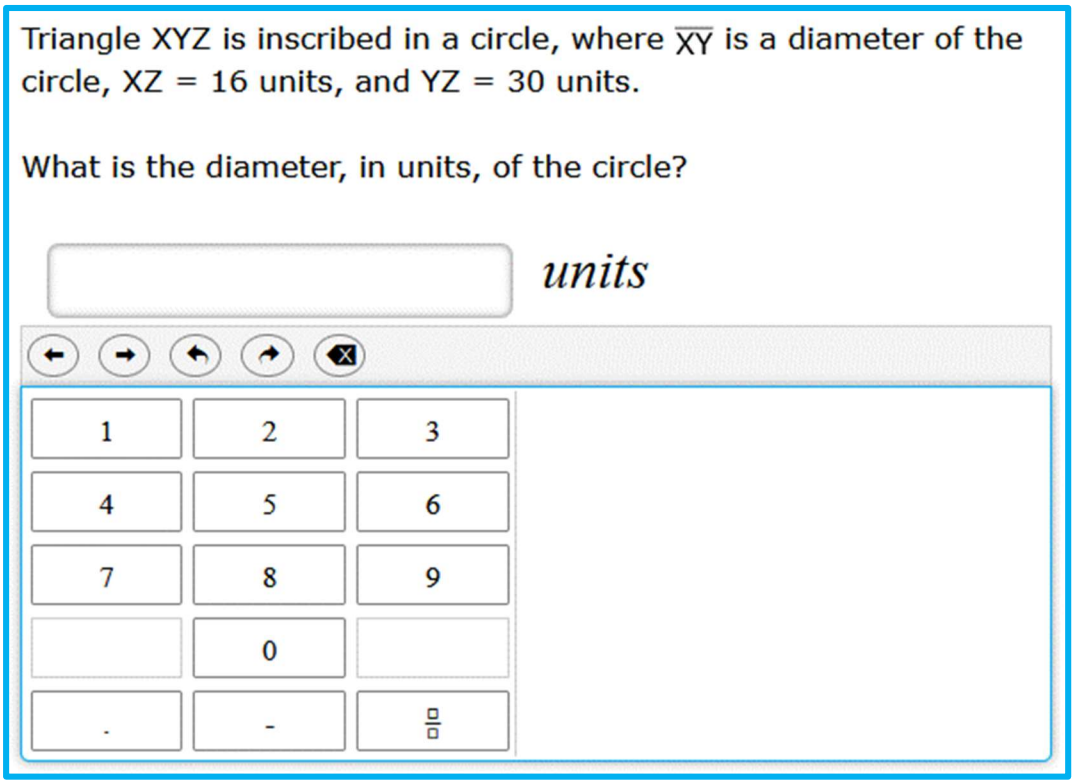
The student may recall the density formula but does not realize the unit on volume is cubic inches instead of inches, and cubes 12.75, $12.75^3 \approx 2072.67$, before using the density formula to find the density to be $\frac{144.585}{2072.67} \approx 0.07 \ \frac{g}{in.^3}$.

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Question 30

Question and Scoring Guidelines

Question 30



Points Possible: 1

Content Cluster: Understand and apply theorems about circles.

Content Standard: Identify and describe relationships among angles, radii, chords, tangents, and arcs and use them to solve problems. Include the relationship between central, inscribed, and circumscribed angles and their intercepted arcs; inscribed angles on a diameter are right angles; the radius of a circle is perpendicular to the tangent where the radius intersects the circle. *(G.C.2)*

Depth of Knowledge: Level 2

Modeling and Reasoning: Yes

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student identifies the correct value, providing evidence of the ability to identify relationships between inscribed angles and diameters and use them to solve problems.

Exemplar Response

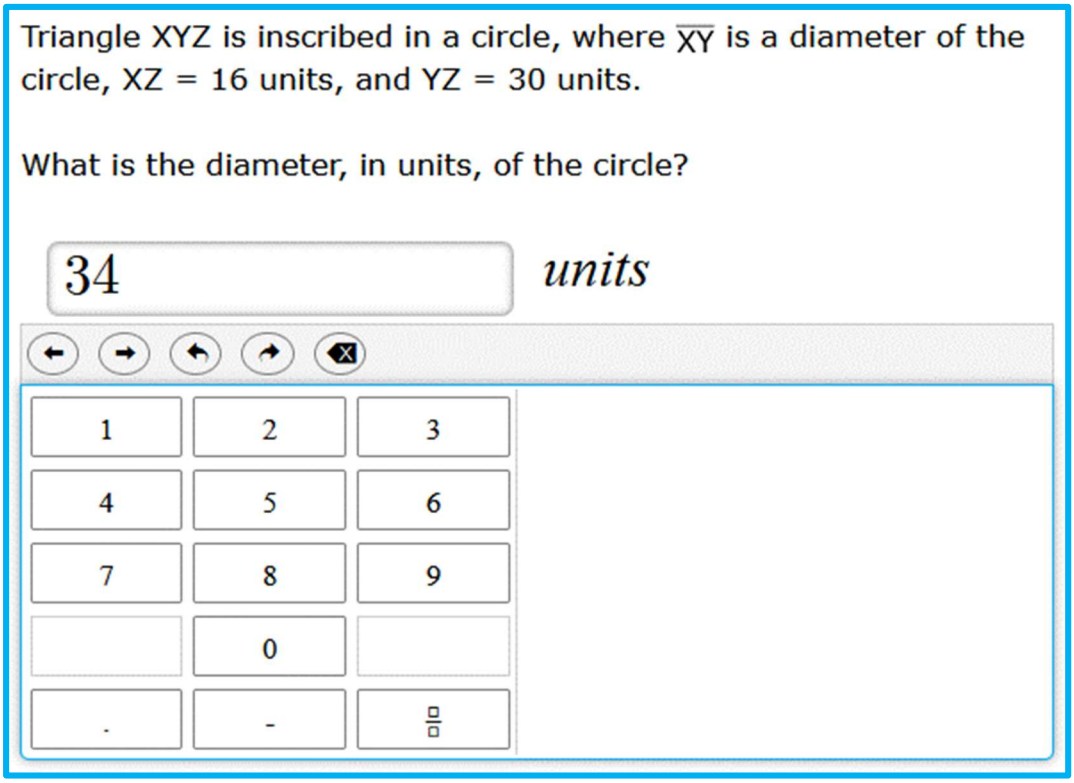
- 34 units
  or
- any equivalent value

Geometry
Spring 2024 Item Release

Question 30

Sample Responses

Sample Response: 1 point



Notes on Scoring

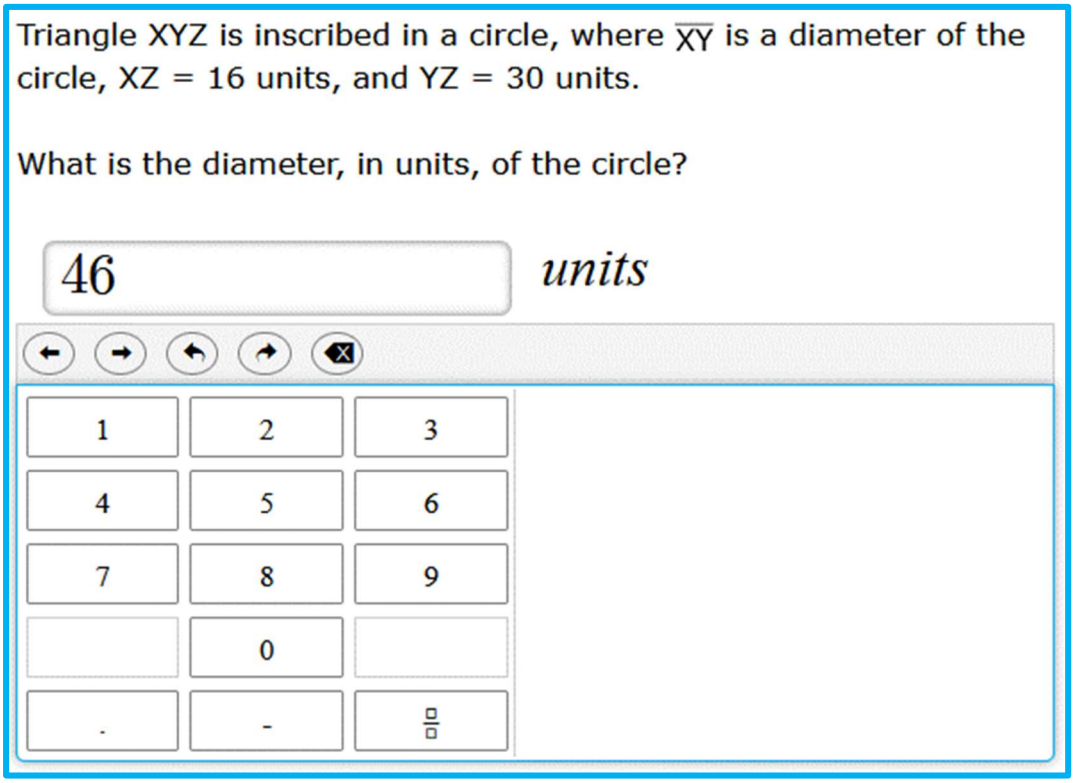
This response earns full credit (1 point) because it shows the correct diameter of the inscribed circle.

The student may understand that since triangle XYZ has a side that is the diameter of the circle, the angle of the triangle opposite this side is 90°, so the triangle is a right triangle. The student can substitute the length of the two legs of the triangle into the Pythagorean Theorem to find the length of the missing side, or the diameter of the circle.

$$\begin{aligned} 16^2 + 30^2 &= c^2 \\ 256 + 900 &= c^2 \\ 1156 &= c^2 \\ 34 &= c \end{aligned}$$

Thus, the diameter of the circle is 34 units.

Sample Response: 0 points



Notes on Scoring

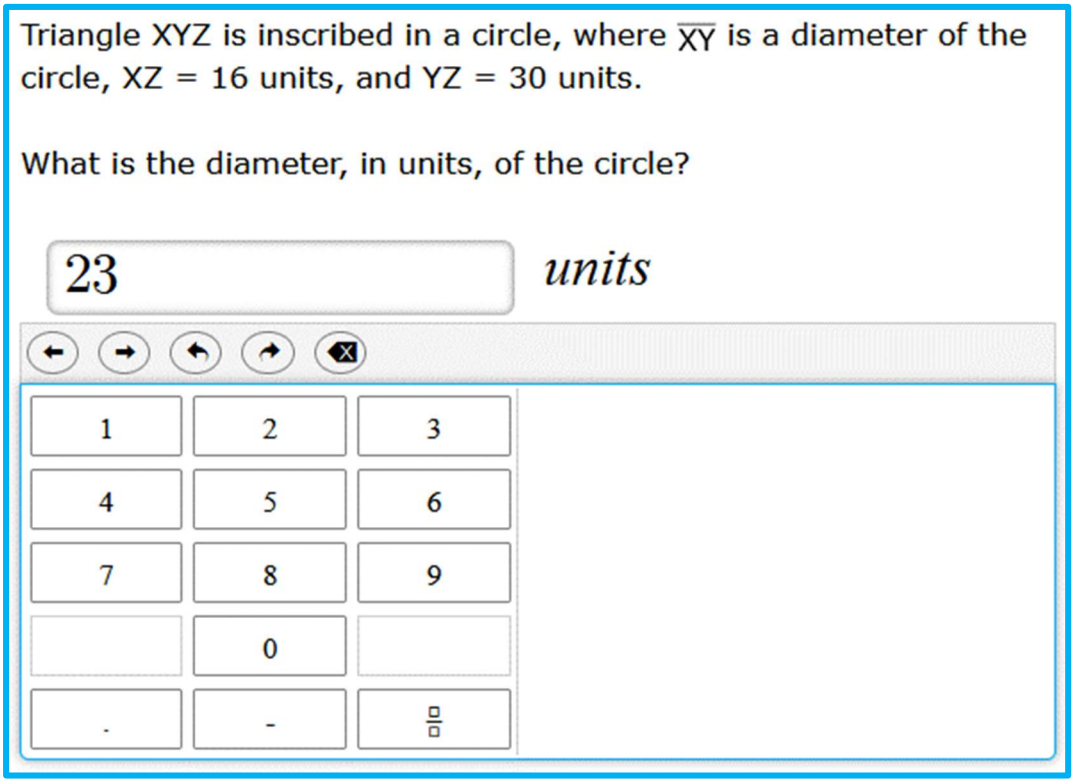
This response earns no credit (0 points) because it shows an incorrect diameter of the inscribed circle.

The student may correctly understand that they can use the Pythagorean Theorem to find the missing side length of the triangle but mistakes the exponents to mean each number is multiplied by 2 instead of multiplied by itself.

$$\begin{aligned} 16^2 + 30^2 &= c^2 \\ 32 + 60 &= c^2 \\ 92 &= c^2 \\ \frac{92}{2} &= 46 \\ 46 &= c \end{aligned}$$

Thus, the student calculates the diameter of the circle to be 46 units.

Sample Response: 0 points



Notes on Scoring

This response earns no credit (0 points) because it shows an incorrect diameter of the inscribed circle.

The student may think that the length of the diameter of the circle is the average of the length of the two legs of the triangle.

$$XY = \frac{(16+30)}{2}$$
$$= \frac{46}{2}$$
$$= 23$$

Thus, the student thinks the diameter of the circle is 23 units.

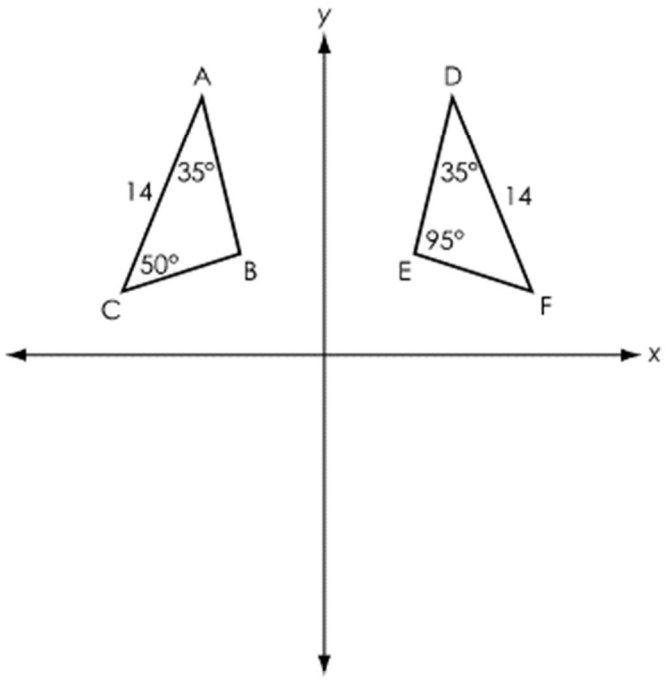
Geometry
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Question 33

Question and Scoring Guidelines

Question 33

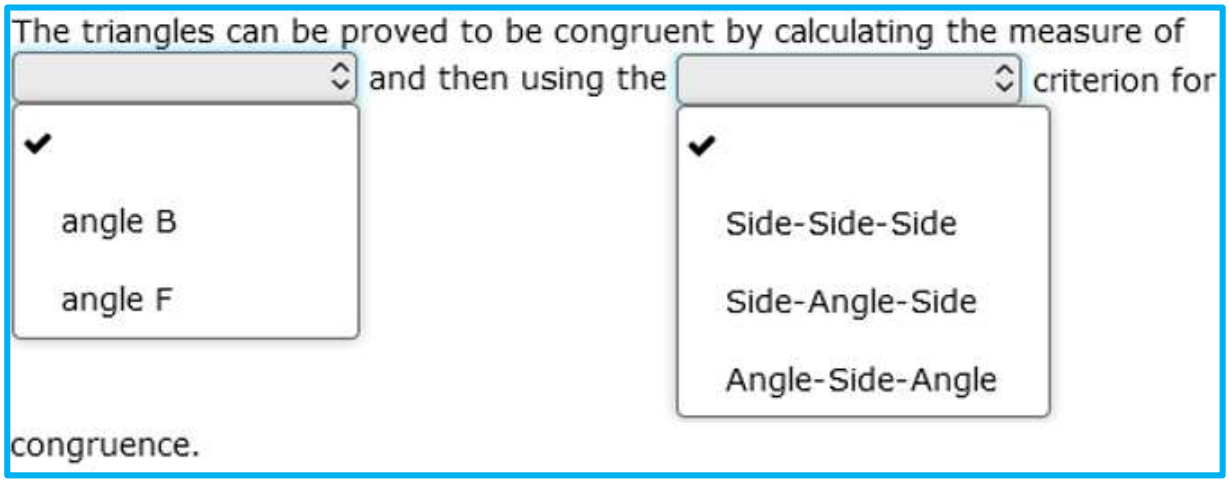
Triangle DEF is the image of triangle ABC after a reflection across the *y*-axis, as shown.



Complete the sentence that can be used to prove the two triangles are congruent.

The triangles can be proved to be congruent by calculating the measure of [ ] and then using the [ ] criterion for congruence.

Drop Down Choices



Points Possible: 1

Content Cluster: Understand congruence in terms of rigid motions.

Content Standard: Explain how the criteria for triangle congruence (ASA, SAS, and SSS) follow from the definition of congruence in terms of rigid motions. *(G.CO.8)*

Depth of Knowledge: Level 1

Modeling and Reasoning: No

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student selects the correct congruence postulate, providing evidence of the ability to prove triangles are congruent.

Exemplar Response

The triangles can be proved to be congruent by calculating the measure of angle F and then using the Angle-Side-Angle criterion for congruence.

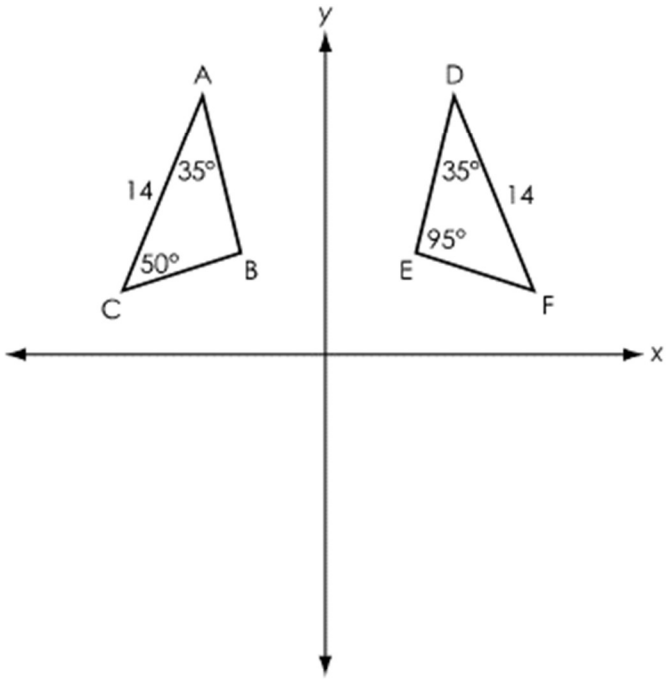
Geometry
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Question 33

Sample Responses

Sample Response: 1 point

Triangle DEF is the image of triangle ABC after a reflection across the *y*-axis, as shown.



Complete the sentence that can be used to prove the two triangles are congruent.

The triangles can be proved to be congruent by calculating the measure of angle F and then using the Angle-Side-Angle criterion for congruence.

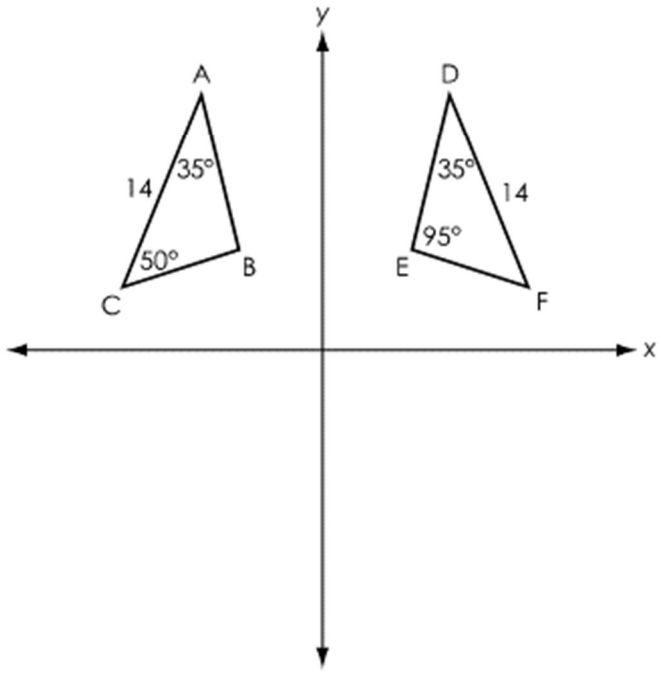
Notes on Scoring

This response earns full credit (1 point) because it correctly completes the statements.

The student may understand that since there is already satisfactory information to use the Angle-Side-Angle criterion for congruence in triangle ABC, they should look in triangle DEF to see what corresponding information is given in triangle DEF to use this criterion to prove the two triangles are congruent. The student identifies that angles A and D correspond and each angle measure is given, that sides $\overline{AC}$ and $\overline{DF}$ correspond and each side length is given, and that angles C and F correspond, but only the measure of angle C is given. Thus, the student identifies if the measure of angle F is calculated, then there is enough information to prove the two triangles are congruent using the Angle-Side-Angle criterion.

Sample Response: 0 points

Triangle DEF is the image of triangle ABC after a reflection across the *y*-axis, as shown.



Complete the sentence that can be used to prove the two triangles are congruent.

The triangles can be proved to be congruent by calculating the measure of angle B and then using the Angle-Side-Angle criterion for congruence.

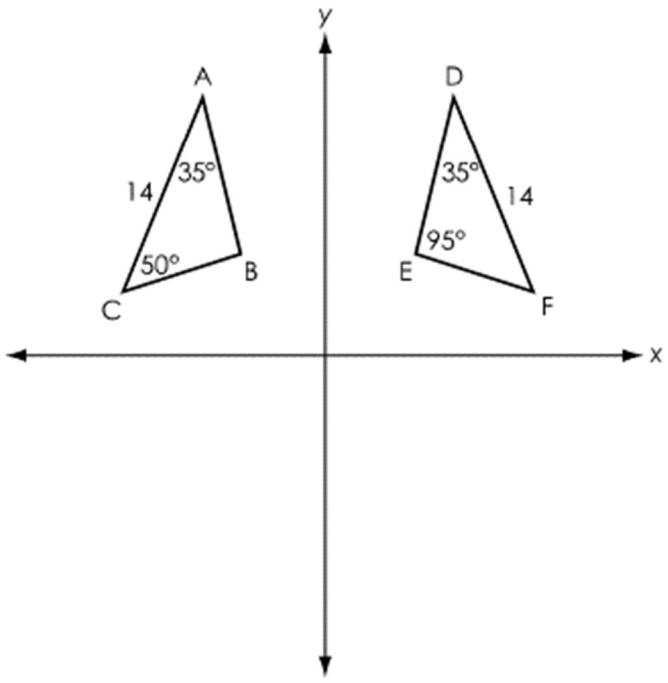
Notes on Scoring

This response earns no credit (0 points) because it incorrectly completes the statements.

The student may have recognized that enough information is given in triangle ABC to potentially prove congruence using Angle-Side-Angle criterion but thought that all the information about the original triangle is necessary when proving congruence instead of observing that the measure of angle F is needed for the Angle-Side-Angle criterion to be used.

Sample Response: 0 points

Triangle DEF is the image of triangle ABC after a reflection across the *y*-axis, as shown.



Complete the sentence that can be used to prove the two triangles are congruent.

The triangles can be proved to be congruent by calculating the measure of angle F and then using the Side-Angle-Side criterion for congruence.

Notes on Scoring

This response earns no credit (0 points) because it incorrectly completes the statements.

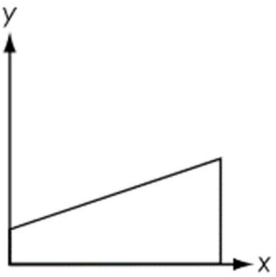
The student may believe that all the angles in the image should be known, thinking angle F must be calculated for that to occur. Understanding that a reflection is a rigid motion, the student identifies that side lengths are preserved as well as angle measures. Therefore, the student identifies that $\overline{EF}$ is congruent to $\overline{BC}$, and thus the triangles can be proven congruent with the Side-Angle-Side Criterion. However, the student does not consider that the congruence of $\overline{EF}$ and $\overline{BC}$ is not established in the given information, and thus only the Angle-Side-Angle criterion can be used to prove the triangles congruent with that information and the addition of the measure of angle F.

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Question 35

Question and Scoring Guidelines

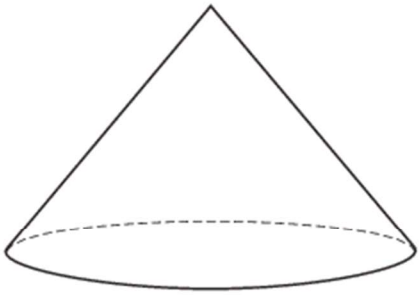
Question 35



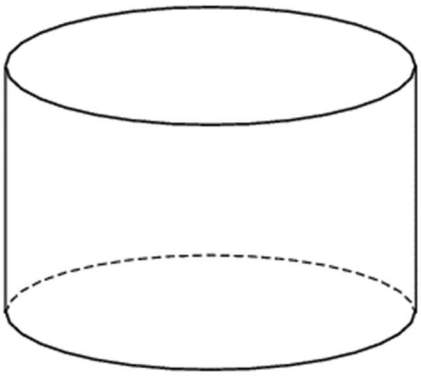
A trapezoid is shown on a coordinate grid. The trapezoid is rotated about the *x*-axis.

What is the resulting shape?

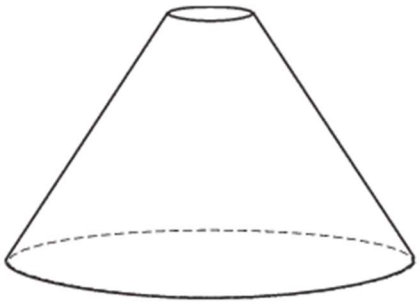
Ⓐ



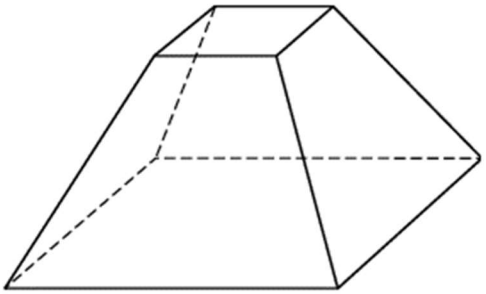
Ⓒ



Ⓑ



Ⓓ



Points Possible: 1

Content Cluster: Visualize relationships between two-dimensional and three-dimensional objects.

Content Standard: Identify the shapes of two-dimensional cross-sections of three-dimensional objects, and identify three-dimensional objects generated by rotations of two-dimensional objects. *(G.GMD.4)*

Depth of Knowledge: Level 1

Modeling and Reasoning: Yes

Scoring Guidelines

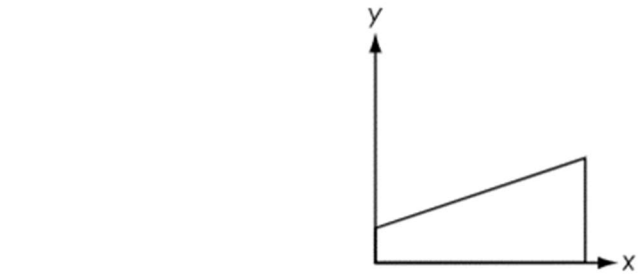
Rationale for Option A: This is incorrect. The student may not consider that the non-vertical linear side of the figure to be rotated is cut by the parallel vertical sides, so it does not cross the x-axis which is necessary to produce the apex of a cone.

Rationale for Option B: **Key** – The student notices that both parallel vertical sides comprising the figure to be rotated would generate two non-congruent circular bases when rotated about the x-axis and that the other side would generate the curved face of a frustum.

Rationale for Option C: This is incorrect. The student may not consider that both parallel vertical sides of the figure to be rotated are different lengths and thus will generate a solid with non-congruent circular bases instead of a cylinder's congruent circular bases.

Rationale for Option D: This is incorrect. The student may think the shape of the figure to be rotated is present in the figure resulting from the rotation, not considering the rotation will generate a curved face.

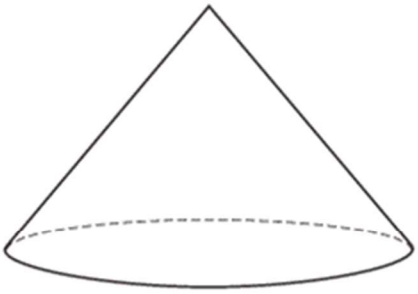
Sample Response: 1 point



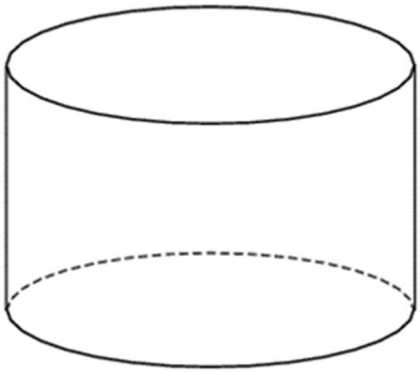
A trapezoid is shown on a coordinate grid. The trapezoid is rotated about the *x*-axis.

What is the resulting shape?

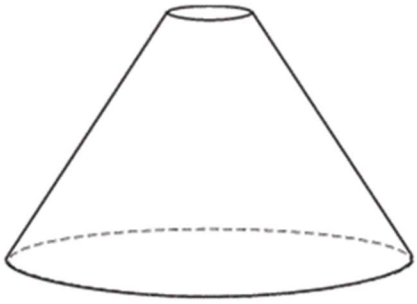
Ⓐ



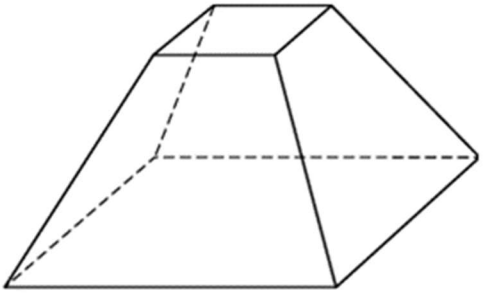
Ⓒ



●



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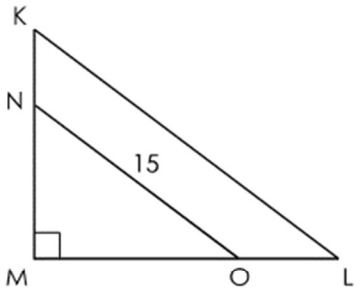
Geometry
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Question 36

Question and Scoring Guidelines

Question 36

Triangle MKL is similar to triangle MNO and NO = 15, as shown.



The sine of ∠MKL is 0.8.

What is the length, in units, of side MO?

units

1 2 3
4 5 6
7 8 9
0
. -

Points Possible: 1

Content Cluster: Define trigonometric ratios and solve problems involving right triangles.

Content Standard: Understand that by similarity, side ratios in right triangles are properties of the angles in the triangle, leading to definitions of trigonometric ratios for acute angles. *(G.SRT.6)*

Depth of Knowledge: Level 2

Modeling and Reasoning: Yes

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student correctly identifies the length, providing evidence of the ability to understand that by similarity, side ratios in right triangles are properties of the angles in the triangle, leading to definitions of trigonometric ratios for acute angles.

Exemplar Response

- 12
  or
- any equivalent value

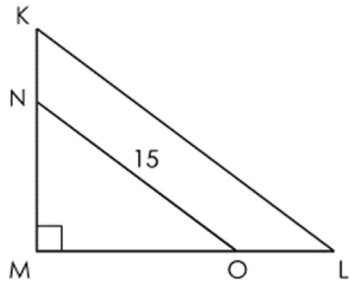
Geometry
Spring 2024 Item Release

Question 36

Sample Responses

Sample Response: 1 point

Triangle MKL is similar to triangle MNO and NO = 15, as shown.



The sine of ∠MKL is 0.8.

What is the length, in units, of side MO?

12 *units*

1	2	3
4	5	6
7	8	9
	0	
.	-	

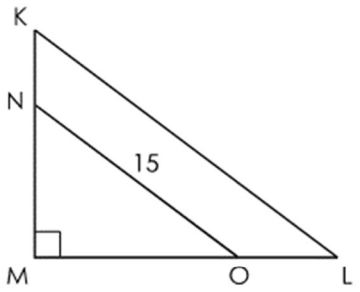
Notes on Scoring

This response earns full credit (1 point) because it shows the correct length of side MO.

The student may understand the sine of an angle is the length of the triangle's side opposite the angle divided by the length of the hypotenuse of the triangle. Since △MKL is similar to △MNO, ∠MKL is congruent to ∠MNO, and the sine of corresponding angles in similar triangles is the same. Thus, sin(∠MKL) = sin(∠MNO), which results in the equation $0.8 = \frac{MO}{15}$. After multiplying 15 to both sides of the equation, the student determines that the length of side MO is 12.

Sample Response: 0 points

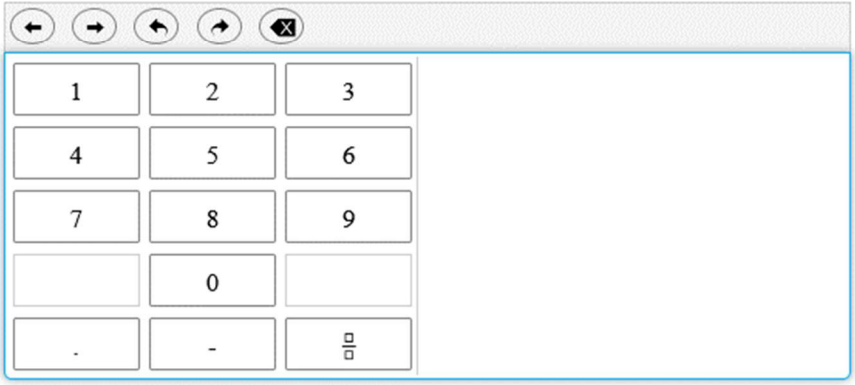
Triangle MKL is similar to triangle MNO and NO = 15, as shown.



The sine of ∠MKL is 0.8.

What is the length, in units, of side MO?

18.75 *units*



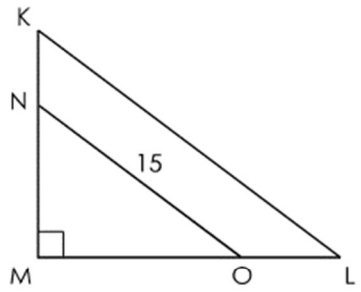
Notes on Scoring

This response earns no credit (0 points) because it shows an incorrect length of side MO.

The student may identify that since △MKL is similar to △MNO, ∠MKL is congruent to ∠MNO and the sine of corresponding angles of similar triangles is the same. However, the student may think $\text{sine} = \frac{\text{hypotenuse}}{\text{opposite}}$ $(0.8 = \frac{15}{\text{MO}})$ instead of $\text{sine} = \frac{\text{opposite}}{\text{hypotenuse}}$ $(0.8 = \frac{\text{MO}}{15})$. The student therefore incorrectly calculates the length of MO to be 18.75 units.

Sample Response: 0 points

Triangle MKL is similar to triangle MNO and NO = 15, as shown.



The sine of ∠MKL is 0.8.

What is the length, in units, of side MO?

9 *units*

1	2	3
4	5	6
7	8	9
	0	
.	-	

Notes on Scoring

This response earns no credit (0 points) because it shows an incorrect length of side MO.

The student may understand that since △MKL is similar to △MNO, ∠MKL is congruent to ∠MNO. However, they may confuse the definitions of sine and cosine, thinking $\text{sine} = \frac{\text{adjacent}}{\text{hypotenuse}}$ $(0.8 = \frac{\text{NM}}{15})$ instead of $\text{sine} = \frac{\text{opposite}}{\text{hypotenuse}}$ $(0.8 = \frac{\text{MO}}{15})$. The student then solves the incorrect equation and determines NM to be 12 units. Finally, to find the length of MO, the student substitutes 12 and 15 into Pythagorean Theorem and solves for the unknown side length with the equation shown.

$$\begin{aligned} 15^2 &= 12^2 + \text{MO}^2 \\ 225 &= 144 + \text{MO}^2 \\ 81 &= \text{MO}^2 \\ \sqrt{81} &= \text{MO} \\ 9 &= \text{MO} \end{aligned}$$

Thus, the student believes the length of MO to be 9 units.

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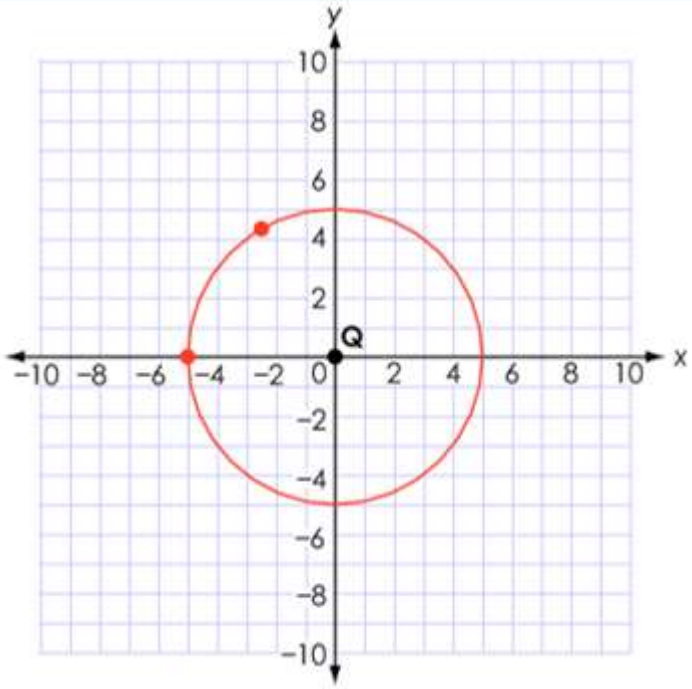
Question 43

Question and Scoring Guidelines

Question 43

Two points of a regular hexagon inscribed in circle Q are shown.

Use the Connect Line tool to complete the hexagon.



Points Possible: 1

Content Cluster: Make geometric constructions.

Content Standard: Construct an equilateral triangle, a square, and a regular hexagon inscribed in a circle. *(G.CO.13)*

Depth of Knowledge: Level 2

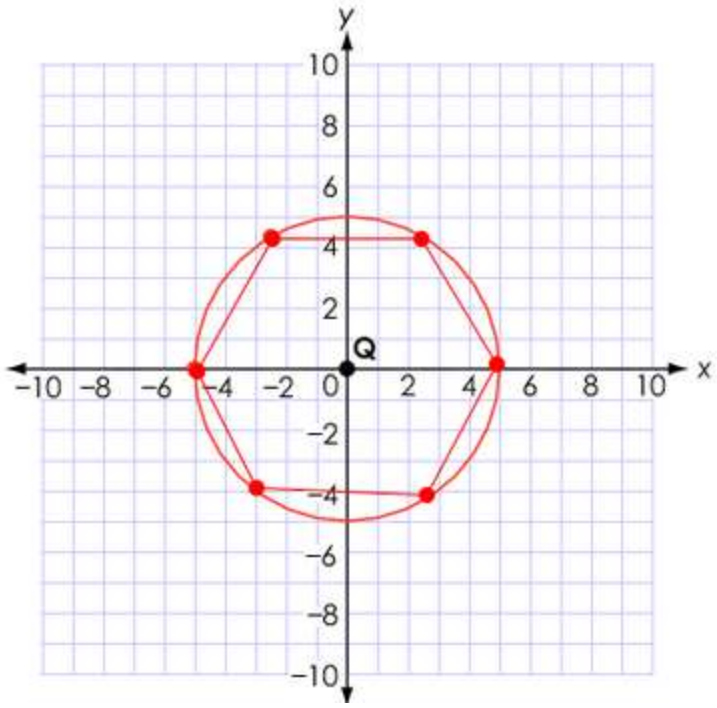
Modeling and Reasoning: Yes

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student draws a correct hexagon, providing evidence of the ability to construct a regular hexagon inscribed in a circle.

Exemplar Response



or

- any line segments connecting non-adjacent vertices within a correct hexagon

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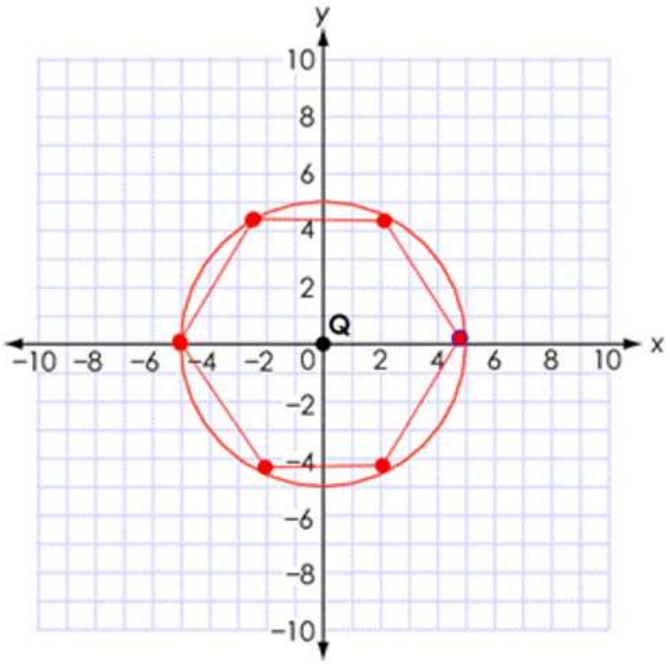
Question 43

Sample Responses

Sample Response: 1 point

Two points of a regular hexagon inscribed in circle Q are shown.

Use the Connect Line tool to complete the hexagon.



Notes on Scoring

This response earns full credit (1 point) because it shows the inscribed regular hexagon.

Since circle Q is centered at the origin, the student may recognize that the inscribed regular hexagon will have symmetry across both the x- and *y*-axis. Using this information, the student plots two vertices, one at $\left(2\frac{2}{3}, 4\frac{1}{3}\right)$ and the other at $(5, 0)$ by reflecting the given vertices, $\left(-2\frac{2}{3}, 4\frac{1}{3}\right)$ and $(-5, 0)$, across the *y*-axis, respectively.

Similarly, the student plots a vertex at about $(-2\frac{2}{3}, -4)$ by reflecting $\left(-2\frac{2}{3}, 4\frac{1}{3}\right)$ across the x-axis.

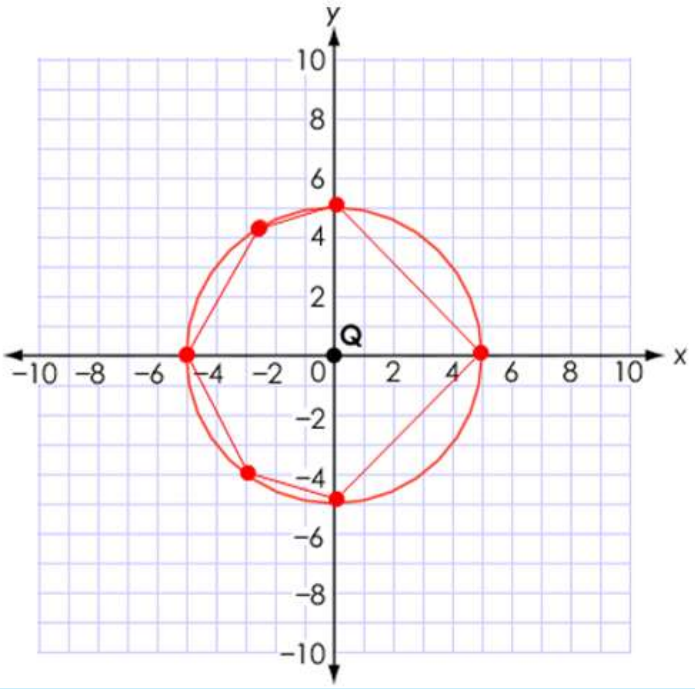
The student then plots the final vertex at about $(2\frac{2}{3}, -4\frac{1}{3})$ by reflecting $(-2\frac{2}{3}, -4)$ across the *y*-axis.

With all six vertices plotted, the student connects each vertex to create the inscribed regular hexagon.

Sample Response: 0 points

Two points of a regular hexagon inscribed in circle Q are shown.

Use the Connect Line tool to complete the hexagon.



Notes on Scoring

This response earns no credit (0 points) because it shows an incorrect inscribed hexagon.

Since circle Q is centered at the origin, the student may identify the inscribed regular hexagon will have symmetry across both the x- and *y*-axis. Using this information, the student plots vertices at the intersections of the circle and the x- and *y*-axes.

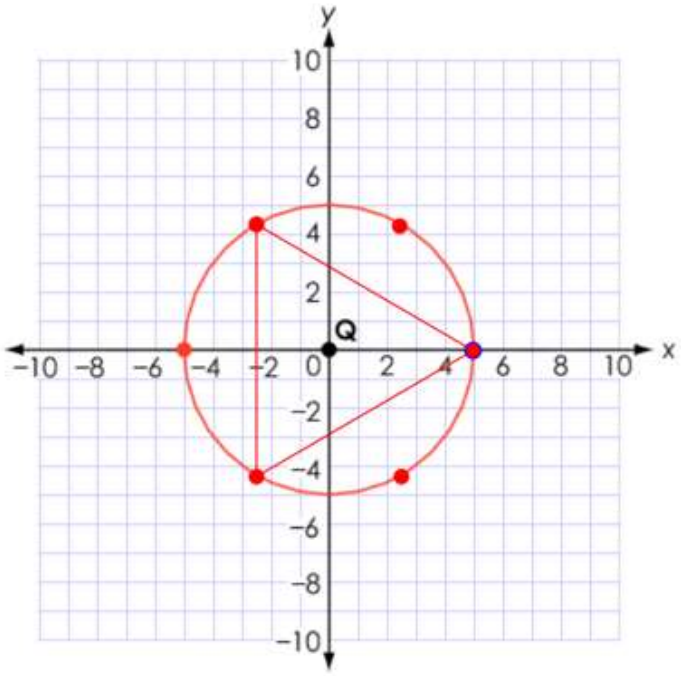
Then, the student reflects the given vertex $\left(-2\frac{2}{3}, 4\frac{1}{3}\right)$ across the x-axis to find the last vertex at $\left(-2\frac{2}{3}, -4\right)$.

With all six vertices plotted, the student connects each vertex to create an inscribed irregular hexagon instead of a regular hexagon.

Sample Response: 0 points

Two points of a regular hexagon inscribed in circle Q are shown.

Use the Connect Line tool to complete the hexagon.



Notes on Scoring

This response earns no credit (0 points) because it shows an inscribed triangle.

The student may correctly plot each vertex of the inscribed hexagon, but ultimately connects every other vertex to create an equilateral triangle instead of a regular hexagon.

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Question 45

Question and Scoring Guidelines

Question 45

A scientist conducts experiments on bacteria using three different conditions, A, B, and C. The table shows the probabilities of the bacteria being affected by each condition.

Probabilities

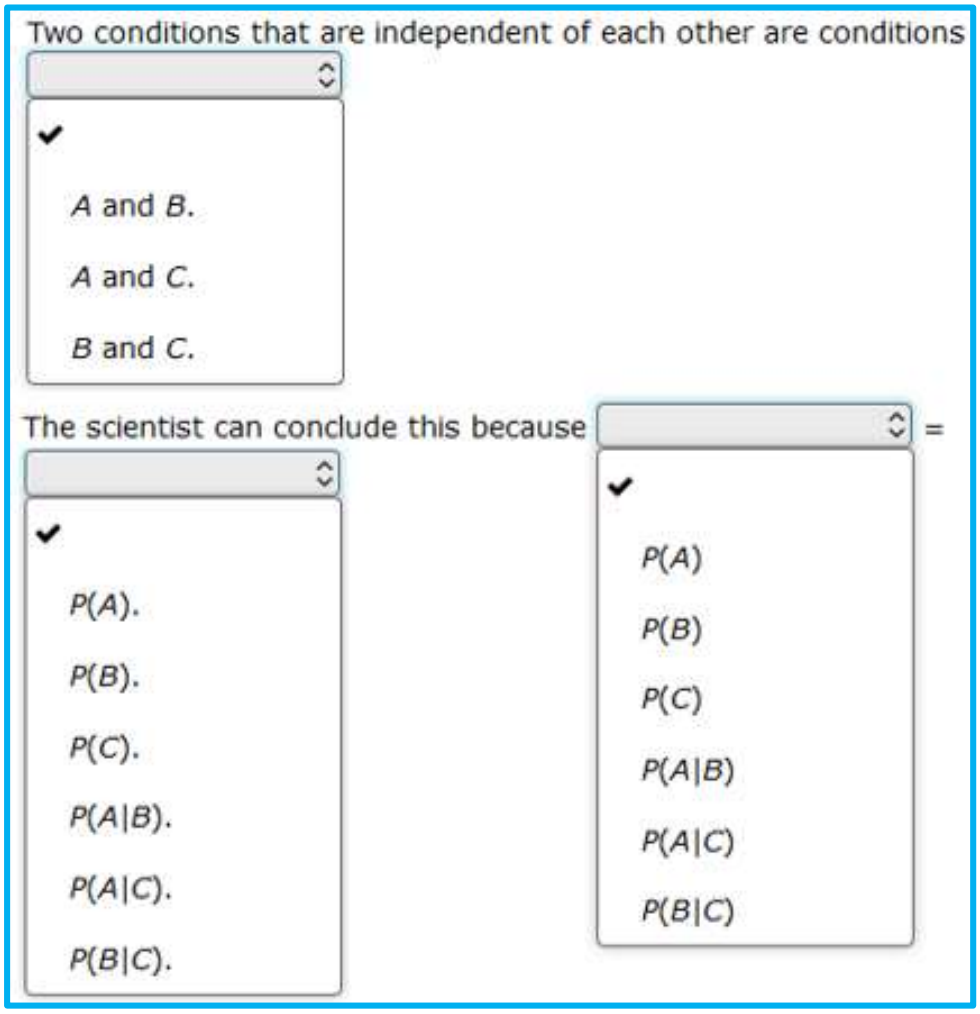
Individual	Conditional
$P(A) = \frac{50}{100}$	$P(A\|B) = \frac{15}{65}$
$P(B) = \frac{65}{100}$	$P(A\|C) = \frac{10}{20}$
$P(C) = \frac{20}{100}$	$P(B\|C) = \frac{15}{20}$

Complete the statements to explain which two conditions are independent.

Two conditions that are independent of each other are conditions [ ]

The scientist can conclude this because [ ] = [ ]

Drop Down Choices



Points Possible: 1

Content Cluster: Understand independence and conditional probability, and use them to interpret data.

Content Standard: Understand the conditional probability of A given B as $P(A \text{ and } B)/P(B)$, and interpret independence of A and B as saying that the conditional probability of A given B is the same as the probability of A, and the conditional probability of B given A is the same as the probability of B. *(S.CP.3)*

Depth of Knowledge: Level 2

Modeling and Reasoning: Yes

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student identifies that conditions *A* and *C* are independent because $P(A) = P(A \mid C)$, providing evidence of the ability to interpret independence of *A* and *C* as saying that the conditional probability of *A* given *C* is the same as the probability of *A*.

Exemplar Response

Two conditions that are independent of each other are conditions [*A* and *C*.]

The scientist can conclude this because [$P(A)$] = [$P(A|C)$.]

or

Two conditions that are independent of each other are conditions [*A* and *C*.]

The scientist can conclude this because [$P(A|C)$] = [$P(A)$.]

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Question 45

Sample Responses

Sample Response: 1 point

A scientist conducts experiments on bacteria using three different conditions, A, B, and C. The table shows the probabilities of the bacteria being affected by each condition.

Probabilities

Individual	Conditional
$P(A) = \frac{50}{100}$	$P(A\|B) = \frac{15}{65}$
$P(B) = \frac{65}{100}$	$P(A\|C) = \frac{10}{20}$
$P(C) = \frac{20}{100}$	$P(B\|C) = \frac{15}{20}$

Complete the statements to explain which two conditions are independent.

Two conditions that are independent of each other are conditions [A and C.]

The scientist can conclude this because [$P(A)$] = [$P(A|C)$.]

Notes on Scoring

This response earns full credit (1 point) because it shows correctly completed statements.

The student may understand that for two conditions, condition 1 and condition 2 to be independent, *P*(Event 1) = *P*(Event 1 | Event 2). The student then reduces the fractions in the given table to compare the individual probabilities of *A* and *B* more easily with the conditional probabilities.

Individual	Conditional
$P(A) = \frac{1}{2}$	$P(A\|B) = \frac{3}{13}$
$P(B) = \frac{13}{20}$	$P(A\|C) = \frac{1}{2}$
$P(C) = \frac{1}{5}$	$P(B\|C) = \frac{3}{4}$

The only individual probability and conditional probability that are equivalent are *P*(*A*) and *P*(*A* | *C*). Thus, the two conditions that are independent of each other are conditions *A* and *C*, and the scientist can conclude that this is because *P*(*A*) = *P*(*A* | *C*).

Sample Response: 1 point

A scientist conducts experiments on bacteria using three different conditions, A, B, and C. The table shows the probabilities of the bacteria being affected by each condition.

Probabilities

Individual	Conditional
$P(A) = \frac{50}{100}$	$P(A\|B) = \frac{15}{65}$
$P(B) = \frac{65}{100}$	$P(A\|C) = \frac{10}{20}$
$P(C) = \frac{20}{100}$	$P(B\|C) = \frac{15}{20}$

Complete the statements to explain which two conditions are independent.

Two conditions that are independent of each other are conditions [A and C.]

The scientist can conclude this because [$P(A|C)$] = [$P(A)$.]

Notes on Scoring

This response earns full credit (1 point) because it correctly completes the statements.

The student may understand that for two conditions to be independent, P(Event 1) = P(Event 1 | Event 2). Correctly comparing the fractions in the table, the student determines that conditions A and C are independent and thus $P(A) = P(A \mid C)$or equivalently $P(A \mid C) = P(A)$.

Sample Response: 0 points

A scientist conducts experiments on bacteria using three different conditions, A, B, and C. The table shows the probabilities of the bacteria being affected by each condition.

Probabilities

Individual	Conditional
$P(A) = \frac{50}{100}$	$P(A\|B) = \frac{15}{65}$
$P(B) = \frac{65}{100}$	$P(A\|C) = \frac{10}{20}$
$P(C) = \frac{20}{100}$	$P(B\|C) = \frac{15}{20}$

Complete the statements to explain which two conditions are independent.

Two conditions that are independent of each other are conditions [A and B.]

The scientist can conclude this because [$P(A|C)$] = [$P(B|C)$.]

Notes on Scoring

This response earns no credit (0 points) because it incorrectly completes the statements.

The student may think that conditional probabilities compared to each other can be used to determine whether two conditions are independent. Considering only the conditional probabilities in the table, the student thinks $P(A \mid C)$ and $P(B \mid C)$ show the two independent conditions since they have the same denominator. Given that C is the common condition in each conditional probability, the student concludes that conditions A and B are independent and that $P(A \mid C) = P(B \mid C)$ overlooking the numerators of each probability.

Sample Response: 0 points

A scientist conducts experiments on bacteria using three different conditions, A, B, and C. The table shows the probabilities of the bacteria being affected by each condition.

Probabilities

Individual	Conditional
$P(A) = \frac{50}{100}$	$P(A\|B) = \frac{15}{65}$
$P(B) = \frac{65}{100}$	$P(A\|C) = \frac{10}{20}$
$P(C) = \frac{20}{100}$	$P(B\|C) = \frac{15}{20}$

Complete the statements to explain which two conditions are independent.

Two conditions that are independent of each other are conditions
A and C.

The scientist can conclude this because $P(A|B)$ = $P(B|C)$.

Notes on Scoring

This response earns no credit (0 points) because it incorrectly completes the statements.

The student may think that two conditions are independent if they have the same conditional probability. The student may notice that $P(A \mid B)$ and $P(B \mid C)$ both have 15 in the numerator, and not realize they have different denominators. Since $P(A \mid B)$ is in the same row as $P(A)$ and $P(B \mid C)$ is in the same row as $P(C)$, the student thinks A and C are independent conditions.

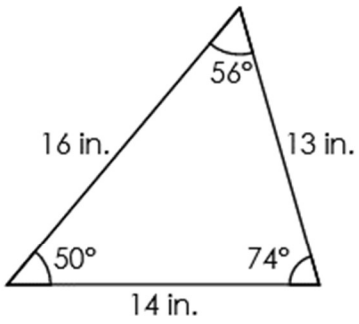
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Question 53

Question and Scoring Guidelines

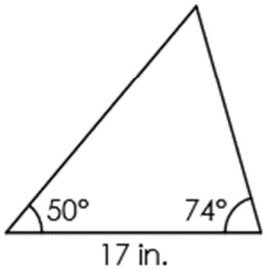
Question 53

A triangle is shown.

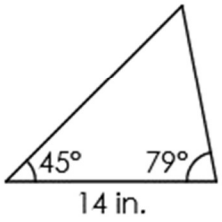


Which triangle is similar?

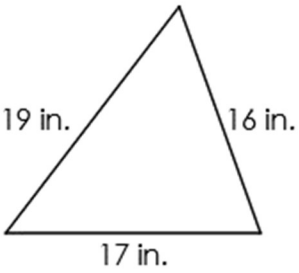
A



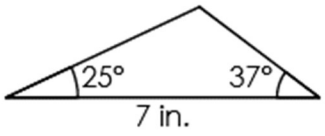
B



C



D



Points Possible: 1

Content Cluster: Understand similarity in terms of similarity transformations.

Content Standard: Use the properties of similarity transformations to establish the AA criterion for two triangles to be similar. *(G.SRT.3)*

Depth of Knowledge: Level 1

Modeling and Reasoning: No

Scoring Guidelines

Rationale for Option A: **Key** – The student recognizes that when two triangles have two pairs of congruent angles, the triangles are similar by the AA criterion, regardless of the length of the side between those two angles.

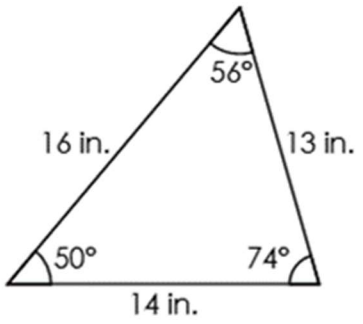
Rationale for Option B: This is incorrect. The student may recognize that the angle across from 14 in. is 56° in both triangles, and may believe that congruence of an angle and the side opposite of it is sufficient to prove similarity.

Rationale for Option C: This is incorrect. The student may think that since every side of this triangle is 3 more than the corresponding side of the original triangle, the sides are proportional, concluding the triangles are similar by the SSS criterion.

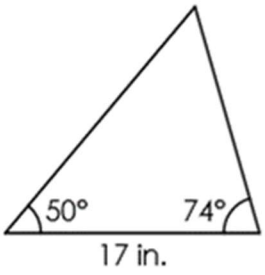
Rationale for Option D: This is incorrect. The student may confuse the ASA criterion to consider the proportionality of the angles and side of a triangle, recognizing that the angles and the side given in this triangle are all $\frac{1}{2}$ times the measure of the corresponding angles and side in the original triangle, concluding this means the triangles are proportional and thus similar by the ASA criterion.

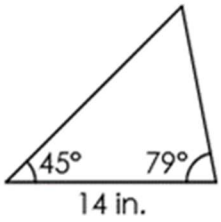
Sample Response: 1 point

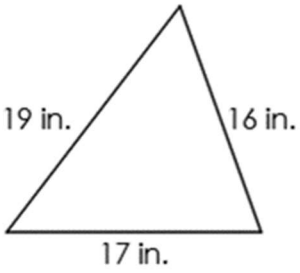
A triangle is shown.



Which triangle is similar?

● 

Ⓑ 

Ⓒ 

Ⓓ 